# Capability Development Document (CDD)

# for

# FAMILY of WEAPON SIGHTS (FWS)

**Increment: I/II**

**ACAT: II**

**Validation Authority: HQDA**

**Approval Authority: HQDA**

**Milestone Decision Authority: PEO Soldier**

**Designation: Joint Information**

**Prepared for: Milestone B Decision**

**Version 1.8**

**September 2011**

**CARDS # 02096**

**Releasability:** Distribution authorized to US Government agencies and their contractors to protect information and technical data that advance current technology or describe new technology in an area of significant or potentially significant military application or that relate to a specific military deficiency of a potential adversary. Information of this type may be classified or unclassified, when unclassified, it is export-controlled and subject to the provisions of DOD Directive 5230.25 (reference c), date of determination: 30 Jul 07. Other requests for this document shall be referred to the: Commander, Maneuver Center of Excellence, ATTN: ATZB-CD, Fort Benning, Georgia 31905-5405.

## EXECUTIVE SUMMARY

The United States Army has identified a requirement for a Family of Weapon Sights (FWS) that shall increase probability of recognition at ranges commensurate with the appropriate weapon system, plus a 20% over match in range of M4/M16 Family, M249, M240, MK19, M2, Shoulder Launched Munitions and Sniper Rifles. A Small Arms Capabilities Based Assessment (CBA) document has identified a decreased ability to recognize and subsequently engage targets, a lowered probability of target suppression or hit from lack of precise target information and an increased risk to the Warfighter. It specifically states a need for a material solution for a weapon sight to enhance the Soldiers ability in acquiring and engaging targets.

In the future, globalization and lower development costs for advanced technologies will provide our adversaries with opportunities to acquire small arms target acquisition capabilities similar to those of U.S. forces today. Future adversaries are also expected to have access to improved Small Arms night vision target acquisition devices. Third-generation Image Intensifier (I2) tubes and second-generation thermal imagers are expected to become the norm for regular forces of some potential adversaries. Additionally there is the potential that adversaries shall have camouflage effective in the infrared spectrum. These developments could enable the threat to seriously challenge our ability to "own the night." Since target acquisition is enabled by various sensors, including the human eye and image intensification devices that are employed with or integrated into Small Arms weapons systems, technological advances in developing these sensors shall increase in significance. Advances in optics technology are of particular interest—maturation of electro-optics/infrared (EO/IR) sensor components and of Aided Target Recognition (ATR) signal-processing schemes. These advancements may also include development of additional specifications and criteria for detection of targets behind opaque materials and with a higher probability of recognition.

This CDD shall address two increments of the FWS, the first increment is the Threshold (T) and the second is the Objective (O) requirement throughout the document. The FWS shall use the evolutionary acquisition approach, delivering the current ground forces the initial increased core capability followed by increasing increments of capability over time as technology matures, culminating with three variants a FWS Individual (FWSI), FWS Crew Served (FWSCS) and FWS Sniper (FWSS). They shall include technology such as, multi-band fused sensors, Rapid Target Acquisition (RTA), advance fire controls for crew served weapons & Sniper variants and clip-on (integration) in-line with Day View Optics.

The FWS shall be used throughout the battlefield by all combat ground forces that routinely dismount and operate in close proximity to combatants. FWS shall be provided to individual Warfighters within Brigade Combat Teams (BCTs) in accordance with the Basis of Issue Plan (BOIP). The current variants of Thermal Weapon Sights (TWS) in the BCT's shall be cascaded to Functional and Multi Function Brigade's in accordance with Army Force Generation (ARFORGEN) cycles. Other Services shall use the FWS requirements as a baseline to develop modifications to best suit their mission sets.

**Table of Contents** **Page**

## POINTS OF CONTACT

| Name | Agency/Organization | Phone Number & DSN | Email Address |
|---|---|---|---|
| COL Geoffrey D. Ellerson | Director, Soldier Requirements Division MCoE | (706) 545-1316<br>DSN 835-1316 | Geoff.ellerson@us.army.mil,<br>Geoff.ellerson@us.army.smil.mil |
| LTC Kyle Korcha | Branch Chief, Electronics and Special Developments Division, Soldier Requirements Division MCoE | (706)545-5069<br>DSN 835-5069 | Kyle.Korcha@us.army.mil,<br>Kyle.a.Korcha@us.army.smil.mil, |
| Mr. John E. Clements | Deputy Branch Chief, Electronics and Special Developments Division, Soldier Requirements Division MCoE | (706) 545-4950<br>DSN 835-4950 | john.e.clements@us.army.mil, |
| Mr. John O'Hara | Combat Developer, Electronics and Special Developments Division, Soldier Requirements Division MCoE | (706) 545-4951<br>DSN 835-4951 | john.p.ohara@us.army.mil,<br>john.p.ohara.ctr@us.army.smil.mil |
| Mr. Tim Boylan | Combat Requirements Analyst, Soldier Branch, MSD, ACDD, ARCIC | (757) 501-5735<br>DSN 501-5735 | tim.boylan@us.army.mil |
| Ms. Moria Ingalls | Optical Engineer PM-Soldier Maneuver Sensors | 703-704-1946<br>DSN 654-1946 | moria.ingalls@us.army.mil |

## Revision History

| Draft Version | Date | Purpose |
|---|---|---|
| 1.0 | 10 January 2010 | Draft Document Developed |
| 1.1 | 31 March 2010 | Developmental (Worldwide) Staffing |
| 1.2 | 13 August 2010 | ARCIC Staffing/Validation (Mr Revell) |
| 1.3 | 2 December 2010 | ARCIC Validated |
| 1.4 | 1 February 2011 | ARSTAF 1-star review |
| 1.5 | 7 April 2011 | ARSTAF 1-star Validated return to gatekeeper |
| 1.6 | 8 July 2011 | ARSTAF 3-star review |
| 1.7 | 8 Aug 2011 | Returned to MCoE for comment resolution |
| 1.8 | 26 Aug 2011 | ARSTAF 3-star Validated return to gatekeeper |

1. Capability Discussion. The purpose of this Capability Development Document (CDD) is to provide the first and second of several increments of the Family of Weapon Sights (FWS) to increase the Warfighter's individual lethality and survivability on the battlefield and provide the Warfighter with the capability to conduct decisive operations under all weather and light conditions without compromising tactical unit integrity. Warfighters and small units acquire (detect, recognize, and identify) threats for engagement with organic or nonorganic fires. Target identification is a process and cognitive task; it is not simply based on what is observed but also includes consideration of the operational context, situational awareness, and applicable rules of engagement. Target acquisition is enabled by various sensors (e.g., human eye, day optics, thermal, image intensification devices and fused multispectral devices) that may be employed with or integrated into Small Arms weapon systems and allow Warfighters to engage targets the sensors enable them to see.

a. The FWS shall increase situational awareness by providing the Warfighter with a capability to detect, recognize and aid in the identification of enemy personnel. The FWS shall help maintain adaptive force dominance with overmatch capabilities in current and future operational environments by providing improved detection and recognition of enemy combatants.

b. The capability gap resides in the inability of the Warfighter to detect, recognize and aid in target identification in all weather conditions both day and night at greater distances than their weapons systems. This system addresses the Joint Operations Concepts (JOpsC) key characteristics through rapid, lethal, precision engagements in adverse conditions; provides endurance and resiliency through improved reliability; supports agility and expeditionary needs through improved maneuverability. Current and future mission sets require the Warfighter to perform various tactical tasks in all conditions while in complex urban, open or mountainous terrain. The gap result is a decreased situational awareness, ability to recognize and subsequently engages targets, a lowered probability of target suppression or hit from a lack of precise target information and culminates in an increased risk to the Warfighter.

c. Operating Environment. U.S. Joint Forces Command's: The Joint Operating Environment JOE - 2008 Challenges and Implications for the Future Joint Force States: Current doctrine and potential commitments range from major theater warfare and close combat in urban and complex terrain to regional instabilities, coalition support, unconventional warfare, foreign internal defense, counter-drug activities, combating terrorism, strategic reconnaissance, direct action, Stability and Support Operations (SASO), peace keeping and peace enforcement, and humanitarian relief efforts. The FWS shall support the range of U.S. Army global operations from Humanitarian Assistance and Peacekeeping, through Counter-terrorism and counterinsurgency, to low intensity conflict and full-scale war. The mission area shall range from villages in developing countries to highly developed urbanized areas. The climatic types shall range from tropic to arctic with weather, vegetation, and terrain conditions associated with these environments.

d. Concept of Employment. The FWS shall be deployed by Soldiers to be used primarily with the small arms; additionally the FWS can be used in the hand held mode. The FWS Individual (FWSI) shall be the primary night vision optic for small arms employment and can be used in conjunction with the day view optic, Rapid Target Acquisition (RTA) sending the reticle and image to the user's Head Mounted Display (HMD) or goggles and in a stand-alone mode. The

FWS Crew Served (FWSCS) shall be used both clip-on or as stand-alone night view optics; The FWS Sniper (FWSS) shall be used in a clip-on in front of the day view optic for all sniper systems and in a stand-alone mode. All variants of the FWS shall facilitate the increase in target interdiction through greater detection, recognition and aiding in identification at greater ranges during periods of darkness and limited visibility.

e. Force Benefits. The fielding of all FWS shall increase lethality and improve situational awareness by extending the range a Warfighter can recognize potential targets. The FWS, used in conjunction with small arms, shall ensure that commanders in the field have constant rapid lethal firepower. The FWS shall have a reduced Size, Weight and Power (SWaP) from the legacy TWS systems, while increasing resolution and Field of View (FOV) over greater ranges.

- The FWSI variant shall be used on the M4 Carbine, M16A2/M16A4 Rifle M249 SAW M136 AT4 M141 Bunker Defeat Munitions (BDM), MK16, MK17 Rifle and M14 offer a RTA capability, in-line night sight capability and a stand-alone sight. This shall enable its use without having to remove the day sight therefore allowing the same aiming Tactics, Techniques and Procedures (TTP)s and eliminating the need for re-zero. The RTA capability enables rapid offensive targeting, by reducing target acquisition and engagement timelines. The FWSI shall benefit the force by increased speed to engagements during Close Quarters Battle (CQB) type engagements without the use of active lasers.

- The FWSCS variant shall offer a night sight capability that mounts on the weapon in place of, or in-line with the day sight, without having to re-zero following detaching and reattaching to the same weapon. A Head Mounted Display (HMD) shall be included to allow for proper firing techniques of the M240 Medium Machine Gun, M2 Heavy Machine Gun (MG), MK19 and MK47 Automatic Grenade Launcher (AGL). It shall be configured as a fused sight with a Low Light Level (LLL) and a thermal sensor. The HMD eliminates the need to obtain eye relief with an optic when firing heavy automatic weapons.

- The FWSS shall be used in a clip-on mode in front of the day view optic for all sniper systems (M24, M110, and M107). Current systems are I2 and don't allow for long distance viewing, range finding with the use of thermal imaging we can achieve distances need for the sniper mission. Repeatability when removing and reinstalling the FWSS shall allow the sniper to use with confidence of zero retention without removing the day view optic.

f. Predecessor Joint Capabilities Integration and Development System (JCIDS) Documents.

(1) Soldier as a System (SaaS) Initial Capabilities Document (ICD), dated 21 Oct 05.

(2) Marine Crew Served Weapon Sight (MCSWS) Initial Capabilities Document (ICD), dated 1 Sep 2004.

(3) Infantry Marine Individual Weapon Sight (IMIWS) Initial Capabilities Document (ICD), dated 1 Sep 2004.

g. Associated Tier 1 and 2 Joint Capability Areas (JCA). The FWS shall support the following Tier 1 through Tier 2 JCAs:

Table 1-1 Associated Tier 1 and 2 Joint Capability Areas (JCA)

| Tier 1 JCA | Tier 2 JCA | FWS Capability(s) | FWS Impact |
|---|---|---|---|
| Force Application | Maneuver<br><br>Engagement | • The ability to effectively engage targets with precision<br><br>• The ability to accurately exact the application of force | • More precise engagement with reduced risk of collateral damage<br><br>• Increase ability to maintain momentum through rapid engagements |
| Command & Control | Understand<br><br>Decide | • Minimize fratricide and collateral damage<br><br>• Detection and location of concealed combatants<br><br>• Avoid potentially hazardous situations | • Operate longer and more effectively<br><br>• Address full range of military operations<br><br>• Increase survivability |
| Battlespace Awareness | Collection | • Avoid potentially hazardous situations<br><br>• The ability to increase individual Warfighter Battle space awareness<br>• Capability to find, identify, and track targets<br>• Detection, location and forwarding of the information | • Enhance force protection<br><br>• Better decisions made faster<br>• Enhance force protection |

2. Analysis Summary. The following Analyses of Alternatives (AOA) was used in supporting the submission and approval of this CDD. The Small Arms CBA confirms the need for a Family of Weapon Sights for the Warfighter to be combat effective while employing a precise Small Arms target acquisition/engagement system. The highlights include the following:

a. FWS Analysis of Alternatives (AOA): DTD 10 Jun 2010

- Individual Rifle Weapon class recommended alternatives are clip-on designs that reduce weapon sight weight and allow the operator to maintain their day capability while providing a thermal channel for no-light and obscured battlefield conditions. The recommended option is a dual-band thermal/low light level tube clip-on system similar in design to the Optical Enhanced Night Vision Goggle (ENVG (O)). The Near InfraRed (NIR) tube has low power and weight impacts, and offers good performance over the effective range of the Individual Rifle. However advances in digital technology could make a digital NIR system a viable option when partnered with the thermal band and would provide additional capability for digital fusion processing.

- Crew Served Weapon Class recommended solutions are standalone designs. However, we desire the FWSCS be used with both Heavy & Medium Machine Guns. The Medium Machine Guns are currently fielded with day optics therefore need to have a clip-on capability. One option is a new LWIR design optimized for range performance given a maximum 10.5cm aperture. The LWIR system is lower power and lighter than the MWIR, but cannot effectively match MWIR range performance without moving to a much larger aperture. Additional options include LWIR/NIR systems based on digital NIR low light level (LLL) technology, with an optically-fused version having power advantages over a digitally-fused version. The NIR channel offers effectiveness improvements (ability to image through glass at shorter ranges and ability to see NIR laser pointers) but has roughly half the range performance of the current Heavy TWS and cannot passively provide imagery in complete darkness.
- Sniper Rifle Weapon Class recommended options for the Sniper Rifle are a trade-off between clip-on and standalone design concepts providing thermal imagery. Clip-on performance is constrained by the interface to the Leopold day optic, and standalone designs require the Sniper to remove the day optic (not desirable). Dual-band clip-on systems are unable to provide adequate range performance within the weight constraints of the AoA, with the thermal channels reaching roughly 85% of the Baseline, and the NIR channels reaching roughly 35% of the baseline.

b. TRADOC Small Arms Capability Based Assessment (CBA) dated: 2 April 2008. Warfighters and small units have a need to detect, recognize, and identify threats for engagement with organic or nonorganic fires. Target identification is a process and cognitive task—it is not simply based on what is observed but also includes consideration of the operational context, situational awareness, and applicable rules of engagement. Target acquisition is enabled by various sensors (e.g., human eye, glass optics, thermal and image intensification devices) that may be employed with or integrated into Small Arms weapon systems and allow Warfighters to fire at targets the sensors enable them to see. The acceptable levels of target acquisition and engagement depend on multiple factors:

- Rules of Engagement (ROE).
- Situational Awareness (SA).
- Optic magnification and resolution.
- Battlefield conditions.
- Weather
- Terrain
- Target signature.
- Field Of View (FOV).

c. Professional Military Experience (PME). Professional Military Experience (PME) is best described as the aggregate and culmination of professional Soldier experience obtained through the synthesis of collective military operations. The PME resulting from Operation Iraqi Freedom (OIF) and Operation Enduring Freedom (OEF) provides empirical evidence that current tactics, techniques and procedures are inadequate in meeting the requirements of the mobile, 24 hours, 7 days a week combat operations environment. It is clear a new capability solution is required that enables the Warfighter effective engagements with confidence through improved detection and recognition of targets at greater ranges during periods of day light, darkness or limited visibility.

3. Concept of Operations Summary. FWS shall support Brigade Combat Teams (BCT) as they engage in stability and security operations against unconventional enemy forces in current operational environments. In the BCT's, the FWS shall be used by unit leaders for surveillance and fire control and by weapons operators to detect, recognize, and engage targets in all visibility conditions. Weapon operators in the BCT shall use RTA portion of the FWSI to survey, detect, select, and engage targets as functions of offensive operations as well as self-protection.

a. Battlespace Awareness. The FWS capabilities shall contribute to Battlespace Awareness, Planning, and Direction with timely target detection and recognition provided to Warfighters. This increases the Warfighter's lethality while addressing the operational gap of rapid situational awareness in support of Joint Operations Concepts (JOpsC).

b. Command and Control. The FWS capabilities contribute to Command and Control with enhanced real time target detection and recognition that enables specific direction and leadership for subordinates in contact. This shared awareness provides enhanced resource synchronization and controlled fires throughout the levels of command with precise and timely Small Arms fires.

c. Force Application. Small units require increasing capability to detect, recognize, and aid in identifying targets for lethal engagement on the battlefield. The FWS improves operations with the ability to influence or disrupt adversarial courses of action. Squad level engagements fix the enemy with precise fires and potentially less collateral damage while maximizing use of covered and concealed fighting positions

d. Operational Mode Summaries (OMS). The wartime OMS is based on a low resolution scenario, which depicts a conflict in the Pacific Command/Combined Forces Command theater of operations. The blue corps consisted of a U.S. mechanized division, a U.S. armored cavalry regiment, a ranger regiment (allied nation), and a U.S. air assault division in a mid-intensity contingency operation to defeat a regional threat force. The scenario depicts the blue corps counterattacking an attacking threat force. The scenario is conventional, with no nuclear or biological operations. Additionally, leveraged the following tactical vignettes from TRADOC Scenario Gist Book 2009; HBCT, Brigade and Below Scenario (BBS), Southwest Asia (SWA) 103.0, SBCT BBS SWA 104.0 and IBCT BBS Northeast Asia (NEA) 10.0. The OMS represents 96 hours of combat that is comprised of four combat operations (attack, defense, movement to contact, and reserve).

4. Threat Summary.

a. Projected Threat Environment. The U.S. shall remain globally engaged in the future, and U.S. forces shall be called on to execute missions across the spectrum of conflict. U.S. forces will conduct forced and early entry operations in environments characterized by urban and complex terrains. Characteristics of these terrains may include insecure flanks, changing enemy tactics and constantly fluctuating situations with the possibility of rapid transitions throughout the spectrum of conflict. The enemy may employ new and advanced optics and munitions in an effort to preclude safe areas for U.S. Forces. To succeed, in addition to conventional forces and attacks, future U.S. forces will have to face Information Operations (IO), likely terrorist attacks, sophisticated ambushes, and a threat that strikes in unconventional and unexpected ways. These forces will have to deal with the key and complex variables of the operational environment, must be prepared to address a full spectrum of military threats, and must counter enemy methods of operation that focus on opportunity and asymmetrical end states. The enemy will be difficult to template as he adapts and attempts to create opportunity. He will develop patterns of operation that shall change as he achieves success or experiences failure in engagements. His doctrine won't change but his way of operating will. There is little likelihood that U.S. forces shall face an enemy who is predictably echeloned in depth, and attempts to destroy us with actions based purely on mass and momentum as most doctrines call for. Instead, potential enemies shall exploit complex terrain and urban environments to obtain tactical advantage and offset our technological and range advantages. He shall conduct decentralized, dispersed, or distributed operations in an attempt to throw U.S. units off balance, but will mass when an opportunity presents itself, or he is forced to mass based on ground maneuver. He shall set sophisticated ambushes and raids using established conventional operations to lure units into kill zones. Finally, the enemy will attempt to blur the lines between combatant and noncombatant personnel on the battlefield. The entire range of conventional and unconventional threats applies to all Family of Weapon Sight platform variants.

b. Threat to Be Countered or Targeted. The Family of Weapon Sights (FWS) will provide U.S. forces the capability to target a range of threats from small bands of insurgents/regional factions groups to major military powers capable of conducting large-scale operations. FWS may be countered by threats that employ camouflage, cover, concealment, denial and deception (C3D2) techniques to include obscurants, decoys, and counter-sensor targeting systems, such as thermal reduction. Also, lasers could have an impact on FWS, the human eye and/or battlefield sensors.

c. Threat to the Capability. It is unlikely that FWS would be specifically targeted by threat fires, but damage could occur from its collocation with other targets and physical destruction could occur from a wide array of threat weapon systems that target the Soldier. Physical threats to FWS, and ultimately the Soldier utilizing FWS, are fragments, bullets, blast, thermobaric, flame, Chemical, Biological, Radiological, Nuclear, and Explosive (CBRNE) weapons and direct energy weapons. Fragment and bullet delivery systems include small arms, grenades, mines, IEDs, VBEIDs, artillery weapons and rockets. Blast weapons include antipersonnel mines, IEDs/VBIEDs, grenades, HE-Frag munitions, and bombs. The Soldier equipped with FWS will always be subject to direct/indirect fire.

d. Baseline Threat Documentation. The Soldier as a System (SaaS) STAR is the baseline document dated 21 Jan 2010. The Future Combat System-BCT STAR, 27 February 2009.

5. <u>Program Summary.</u>

a. The FWS development program is an evolutionary acquisition strategy developed in conjunction with the Product Manager, Soldier Maneuver Sensors. The Advanced Weapon Sight Technology (AWST) program being conducted by the Night Vision Labs will be the RDT&E effort used to produce designs for a Family of Weapon Sights (FWS), and fabrication of select technology demonstrators using advancements in imaging, detection, and processing. FWS Increment I and II shall capitalize on the latest available night vision technologies to provide a lightweight, low cost, night or limited visibility weapons optic for the Soldier. FWS leverages lessons learned, hardware, and software from existing LLL, and Thermal Weapons Sight (TWS) programs to the maximum extent possible. FWS shall provide the mounted or dismounted Soldier with the capability to employ Small Arms at greater ranges with improved clarity. FWS shall be employed with the following weapons: M16A4/M4 Modular Weapon Systems, MK16, MK17, M14, M249 Squad Automatic Weapon, M240 Medium Machine Gun, MK 19 Grenade Launcher, MK47 Striker Automatic Grenade Launcher, M2 Heavy Machine Gun, M107, M24, M110 Sniper Rifles and M136 AT4 and M141 BDM using Mil-Std-1913 Picatinny rails.

b. Other Services may use the FWS requirements as a baseline to develop modifications to best suit their mission sets. The expected system life for the FWS is 20 years with Low Rate Initial Production planned for Q2FY15 with a First Unit Equipped (FUE) of Q2FY17.

6. <u>System Capabilities Required for the Current Increment.</u>

a. Statutory and Compliance KPPs not appropriate for FWS.

(1) Net Ready-Key Performance Parameter (NR-KPP) - This capability does not have a command, control, communications, computers, and intelligence, surveillance, and reconnaissance (C4ISR) interface with any other system or capability. This, along with the development of the operational architecture products, supports the conclusion that system and technical views, as well as the full development of an NR-KPP appendix, is not applicable. Therefore, a NR-KPP is not appropriate for the FWS.

(2) Survivability - FWS is not a manned system (as in tank, plane, ship, HMMWV, etc). Therefore, the Survivability KPP is not appropriate for FWS.

(3) Force Protection - The FWS is not designed to directly prevent or mitigate hostile actions against personnel. Therefore, there is no Force Protection KPP included in this document.

(4) Sustainment - The sponsor has determined FWS sustainment to be not applicable/appropriate as a KPP in accordance with provisions documented in Manual for the Operation of the Joint Capabilities Integration and Development System, 31 July 2009, Enclosure B (Performance Attributes and Key Performance Parameters), paragraph 2b. Accordingly, the operational availability component of FWS sustainment is addressed as a KSA, together with materiel reliability and ownership cost, given consideration that sustainment for FWS leverages lessons learned from the predecessor TWS program and thereby shall continue to utilize the maintenance and support concept implemented (in-place) for that system.

(5) Energy Efficiency -This KPP only applies to emerging platforms, systems and subsystems that obtain their power either directly or indirectly from hydrocarbon energy sources. The FWS does not obtain power from a platform system therefore, this KPP is not appropriate for FWS.

(6) System Training -. It was determined through analysis that System Training should not be a KPP; FWS training will be incorporated into existing training and education. Training is not a significant part of the total life cycle cost.

b. Key Performance Parameters (KPPs) for the Family of Weapon Sights. *The KPP Table is located in Appendix E.*

(1) KPP 1  The FWSI shall mount in line with the M150 Rifle Combat Optic (RCO) and the M68 Close Combat Optic (CCO) on individual Soldier weapons and shall have unity magnification. It shall also have the ability to operate in a standalone configuration. This design shall include a Rapid Target Acquisition (RTA) module which displays the zeroed reticle and a portion of the thermal image in the HMD or goggle. The RTA design shall include a wired (T) wireless (O) interface between the FWSI and HMD or goggle and the associated signal processing electronics. FWSI may have a range finding capability to the effective ranges of the weapons described in this CDD within of +/- 5 meters (O). This sight shall also be a standalone limited visibility/night sight for the M16A2,A4/M4/M249, M136 AT4 and M141 BDM (T) and MK16, MK17, M14(O). Soldiers shall also be able to shoulder the weapon and obtain proper sight picture through the day view optic, and clip-on thermal portion of the system, this allows for more stable, longer shots. The FWSCS must be a clip-on (in-line with day view optic) or standalone limited visibility/night sight and mount on the MIL-STD-1913 Integrated Rail so as to be compatible with the M240, M2, MK19, MK47. The FWSCS shall have the ability to interface with a laser range finding device and receive range information as inputs to ballistic equations embedded within the FWSCS (O). In the clip-on mode it must be able to align with the Machine Gun Optic M145 (MGO). The FWSCS shall include a HMD Wired (T) Wireless (O). FWSCS shall have a range finding capability of +/- 5 meter (O) and wind measurement +/- 0.23 meters per second (O). FWSS shall mount on the MIL-STD-1913 Integrated Rail of the M110, M24 and M107 in-line with the Leupold MK4 4.5-14X day optic. It must be capable of being used in the stand-alone mode. It shall not interfere in any way with the normal operation or inherent accuracy of the host weapon. All variants shall allow for detach and reattaching to the same weapon with a constant zero repeatability within .50 mrad (T) .25 mrad (O). Additionally, all variants shall provide the proper form, fit, and functionality to be removed from the host weapon and used in the hand-held mode. FWSS shall have a range finding capability of +/-1 meter (O) and wind measurement +/- 0.23 meters per second (O).

**Rationale:** The FWS requires a common mounting interface that shall allow operational mounting without a decrease/loss of zero alignment. The RTA capability shall facilitate offensive target acquisition for the Soldier by displaying an image and zeroed reticle to the HMD or goggle. Additionally, following dismount and remount zeroed optics with little to no loss of zero allows for the use of other fire control systems (e.g. DVO, night vision) to address changing METT-TC requirements. The FWSCS must be both stand-alone because there is no day optic for the M2 and MK19 weapons systems and Clip-on that allows for the in-line use with a MGO while mounted on the M240 MG. While in the stand alone mode it FWSCS will provide an adjusted reticle which shall allow the Soldier to engage distant targets without having to calculate distance or adjust aim point for range. It shall also have a HMD in order to maintain a thermal displayed picture in various user firing positions. Leaders indicate the need to use their sights in a hand-held mode for reconnaissance, situational awareness and command and control missions support.

(2) KPP 2 Target Detection during night engagements by a trained Soldier employing their assigned weapon system must be able to recognize a personnel-sized target (standing) at the following ranges 70% of the time: FWSI 960 meters (T) and 1200 meters (O), FWSCS 2400 meters (T) 2600 meters (O), FWSS 1800 meters (T), and 2200 meters (O). Target Detection during limited visibility as in fog, salt fog, haze, mist, snow, dust, smoke and other battlefield obscurants like white phosphorous (WP), hexachloro-ethane-zinc oxide (HC) by a trained Soldier employing their assigned weapon system must be able to recognize a personnel-sized target (standing) with such clarity it enables a correct engagement decision at the following ranges 90% of the time: FWSI 300 meters (T) and 480 meters (O), FWSCS 500 meters (T) and 600 meters (O), and FWSS 600 meters (T), and 800 meters (O). Modeling standard used for cycles on target will be determined using the most current Detect, Acquire, Recognize and Identify (DARI) observer test.

**Rationale:** The capability to recognize and subsequently engage threat personnel at night and/or limited visibility conditions supports Warfighter TTP's. Soldiers must be able to acquire targets and then use different levels of recognition before they can engage. The ability to recognize threat personnel at range supports precision fires and inherently provides minimal margin of error.

(3) KPP 3 The FWSI shall have an 18$^{o}$ (T) and 26$^{o}$ (O) Horizontal Direct FOV. The FWSCS shall have a 9$^{o}$ (T) 18$^{o}$ (O) Horizontal Direct FOV. T=O. The FWSS shall have a 4$^{o}$ (T) and 9$^{o}$ (O) Horizontal Direct FOV. T=O. This is measured while in the stand alone mode.

**Rationale:** The wider FOV creates lower resolution but shall support scanning and surveillance, the narrow FOV provides a higher resolution image that supports target recognition at longer ranges. The wider FOV of the FWSI is to support Soldiers in the close-in thru the mid-range battle. The narrow FOV for the Crew served and Sniper variants shall allow for a longer range during surveillance and target identification and interdiction operations.

(4) KPP 4 Weight. The total system weight of the FWSI and batteries to include additional loaded battery cassette, without carrying case cannot exceed 2.0 lbs (T) 1.75 lbs (O). RTA processor shall weigh less than 2.0 lbs (T) 1.5 lbs (O) excluding sight and HMD. The total system weight of the FWSCS and batteries to include additional loaded battery cassette, without carrying case cannot exceed 3.25 lbs. (T) 2.75 lbs (O). The total system weight of the FWSS and batteries to include additional loaded battery cassette, without carrying case cannot exceed 2.0lbs (T) 1.75 lbs (O). The size of the FWS shall be such as to not interfere with the use of the host weapons.

**Rationale:** Warfighters are often deployed on long duration missions with no chance of resupply. They are already carrying exceptionally heavy loads and it is imperative the FWS does not add significant weight to this load.

(5) KPP 5 Power. All FWS variants shall be operable by using a commercially available battery compliant with Communication Electronics Command (CECOM's) Preferred Power List at temperatures between -40ºC and +49ºC with an allowance of no greater than 40% reduction in battery life and efficiency between -20ºC and -40ºC for at least 7 hours full-on with one battery change (T). All FWS variants shall be operable by using a commercially available battery

compliant with CECOM's Preferred Power List at temperatures between -46ºC and +49ºC with an allowance of no greater than 40% reduction in battery life and efficiency between -20ºC and -46ºC for at least 7 hours using one set of batteries. (O) All FWS variants shall meet requirements under hot, basic, and cold as defined in AR 70-38 (T=O). All FWS variants shall include a low battery indicator in the field of view that alerts the operator when approx 15 minutes of battery life remains (T). All FWS variants shall include a battery life indicator in the field of view that provides battery status throughout the battery life (O).

**Rationale:** Using common batteries that are approved by CECOM, commercially available and/or within the military inventory shall reduce the cumulative number of batteries carried by the Warfighter. Warning for failing power or diminished capacity prevents Soldiers from making battery changes during mission.

c. Key System Attributes (KSAs). *The KSA Table is located in Appendix E.*

(1) KSA 1 FWSI shall have a minimum focus of 5 meter, (T) 1 meter (O) and FWSCS and FWSS shall have a focus range of 10 meters to infinity (T) 5 meters to infinity (O) and shall maintain boresight through the focus range.

**Rationale:** This shall ensure both the close fight and medium to long range engagements are supportable by the optics. The use of the term infinity is an industry standard when referring to focusing of optics.

(2) KSA 2 All FWS variants shall not emit noise that is detectable by the average human ear at a distance beyond 5 meters in any direction during use. When the user is not wearing protective eyewear or goggles, the FWS all variants shall not emit stray light detectable by an unaided eye beyond 5 meters (T=O).

**Rationale:** To prevent disclosure in situations such as night ambush or listening/observation posts where the enemy may be allowed to approach very close to the operator's position, or the Soldiers are moving in close proximity to their enemy. Noise and light discipline is basic Soldier skills for survival. Additionally, Soldiers have a need to listen to their surroundings to aid in survivability.

(3) KSA 3 The total cost of FWSI (to include: basic unit, hard and soft carrying cases, all ancillary equipment (includes initial fielding of 2 sets go to war batteries with cassettes, 4 sets rechargeable batteries and charger, necessary brackets) and other sustainment costs for the life of the product shall not exceed $23,020 per unit (T) and $25,320 per unit (O) in BY11 dollars

(a) The total cost of FWSCS (to include: basic unit, hard and soft carrying case, all ancillary equipment (includes initial fielding of 2 sets go to war batteries with cassettes, 4 sets rechargeable batteries and charger, necessary brackets) and other sustainment costs for the life of the product shall not exceed $23,600 per unit (T) and $25,960 per unit (O) in BY11 dollars.

(b) The total cost of FWSS (to include: basic unit, hard and soft carrying case, all ancillary equipment (includes initial fielding of 2 sets go to war batteries with cassettes, 4 sets rechargeable batteries and charger, necessary brackets) and other sustainment costs for the life of the product shall not exceed $23,150 per unit (T) and $25,460 per unit (O) in BY11 dollars.

**Rationale:** Total life cycle cost to procure & maintain FWS within the force is to be minimized to the extent practical while maintaining an efficient support structure within the force. The total ownership costs are estimates provided by the PM and will be adjusted throughout the life of the program. (Set of batteries is defined as total number of batteries to fill battery cassette).

(4) KSA 4 Operational Availability ($A_o$) of the FWS will be no less than 81 percent (T), 90 percent (O) measured over the 8-day wartime usage period (consisting of two aggregate 96-hour wartime scenarios) depicted in the Operational Mode Summary/Mission Profile (OMS/MP), where system abort failure events as described by the FWS Reliability Failure Definition and Scoring Criteria (FDSC) are the primary indicator of equipment not ready/available conditions.

**Rationale:** Consistent with the recurring wartime usage cycle depicted in the OMS/MP (8 days aggregate system usage and 4 days of stand-down time), the threshold $A_o$ supports mission accomplishment at the unit level by providing assurance that core fighters/personnel within the platoon (riflemen, automatic riflemen, and machine gunners, as well as team leaders) will have functional day/night and adverse weather small arms weapon sighting capability operationally available to them throughout conduct of the 8-day wartime operating pulse. Threshold requirement reflects (accounts for) TWSs within the platoon issued to personnel in squad and platoon leader roles as providing a backup capability to overcome mission critical FWS failures, enabling core fighters/personnel to retain thermal sighting capability through redistribution/reallocation of the TWSs, as necessary, to fill any voids created by inoperable FWSs (in such cases the affected squad/platoon leader(s) who redistribute their TWS will still have residual day and night target engagement / surveillance capability using their day optic (plus backup iron sights) and infrared aiming light - night observation device combination). Objective $A_o$ reflects capability which aligns with that desired for Army Brigade Combat Team Modernized units. $A_o$ characterization within the context of requirements derivation analysis is based on system usage involving 95.1 FWS operating hours for the 8-day pulse consistent with the wartime OMS/MP, a maintenance support concept comparable to that utilized for the current TWS the FWS will replace/displace (in which field level repair is limited to the replacement of easily accessible/replaceable components such as knobs, eyecup, mount, objective lens cover, etc., and repairs entailing off-system maintenance are performed by the Intelligence and Electronic Warfare Regional Support Center), the projected average repair cycle administrative and logistics delay time of 195.0 hours incurred per system abort failure event (determined from the weighted average of median repair cycle time values for predecessor thermal sighting equipment utilized by units operating in the Iraqi theater during fiscal years 2004 through 2009, as obtained from the LOGSA Logistics Information Warehouse data base), and the logistics attribute requirement for FWS maximum time to repair, where all repair actions for failures not resulting in an FWS system abort are deferred until the operating pulse has been completed).

(5) KSA 5 The FWS will achieve/demonstrate an operational (probabilistic) reliability of not less than .90 (T), .95 (O) related to successfully completing the 96-hour wartime scenario depicted by the OMS/MP without incurring an essential function failure (EFF) that results in a system abort (SA) as defined in the Reliability FDSC. The FWS operational reliability required

to complete the 96-hour scenario without incurring an EFF of any severity will be no less than .84 (T), .92 (O).

Rationale:  RAM analysis based on the FWS wartime OMS/MP, basis of issue, and a support concept comparable to that employed for the current thermal sight units the FWS will replace/displace (inclusive of the associated repair cycle delay time) indicates the threshold reliability requirement associated with SA failure events (expressed operationally as probability values in accordance with current TRADOC guidance) will enable the FWS to achieve the requisite threshold $A_o$ throughout the conduct of mission operations; objective reliability reflects the capability needed to meet the $A_o$ objective (threshold and objective operational reliability requirements correlate to 458 and 909 operating hours Mean Time Between System Abort, respectively, when assessed in conjunction with usage factors defined in paragraph 2 of the OMS/MP (47.55 FWS operating hours for the 96-hour wartime scenario)).  Reliability requirements associated with EFFs address the impact of reliability and associated essential maintenance demands on logistics relative to maintaining steady state $A_o$ during a protracted (extended) period of operations consistent with the annual usage cycle defined in paragraph 8.7 of the OMS/MP.  Threshold EFF reliability requirement reflects the supporting capability needed to retain at least 81 percent of the FWSs in an operationally available status on a steady state (e.g., annualized) basis, whereas the objective reliability reflects the capability needed to retain 90 percent steady state availability (threshold and objective EFF-related operational reliability requirements correlate to 275 and 581 operating hours Mean Time Between Essential Function Failure, respectively).

d. Additional System Attributes.  *The Additional Performance Attributes table is located in Appendix E.*

(1) Attribute 1 Carrying Case.  The FWS shall have a soft water resistant carrying case that shall attach to individual load bearing equipment.  The case shall provide space for the sight, one M16A1/A2 mounting bracket, one spare loaded battery cassette, operator's manual, lens cleaning materials, and quick reference card.  A hard transit case shall also be provided for the FWS (T=O).

**Rationale:** A soft protective carrying case shall be necessary for normal storage and transport. A hard protective case shall be required for transit and long term storage.

(2) Attribute 2 Battery Removal: The battery cassette and batteries must be easily replaced without using tools, without removing the sight from the weapon, and without affecting zeroing/sight alignment (T=O).

**Rationale:** Soldiers are changing batteries during missions and must be able to accomplish this task quickly and without tools.

(3) Attribute 3 Compatibility with Personal Protective Equipment (PPE): All FWS systems shall be capable of being operated while wearing issued personnel protective equipment.  Note: PPE includes equipment such as the Light Weight Helmet, MICH Helmet, Modular Tactical Vest, Gloves (cold weather, NOMEX,) Protective Eyewear, and MOPP IV gear. (T=O)

**Rationale:** The FWS shall be used in varying combat situations that may require many varieties of clothing and equipment types. Soldiers should not be limited in PPE usage by the FWS sight.

(4) Attribute 4 Reticle: Each FWS shall have internal reticles that are compatible with each supported weapon. All FWS reticles will have boresighting capability, which requires no special tools to operate. These reticles shall be selectable by the operator (T=O).

**Rationale:** When in the stand alone mode the FWSI, FWCS and FWSS won't have the day optics reticle for ranging and engaging targets, therefore needing a selectable reticle to support all intended weapon variants is required.

(5) Attribute 5 All FWS variants should allow the operator to control and adjust the display brightness and gain controls (T=O).

**Rationale:** This capability enables the Soldier to adjust the brightness of the display for both day and night operations and adjust the gain of the thermal image for low or high contrast thermal conditions.

(6) Attribute 6 The FWS materials shall protect it from corrosion in all environments and weather conditions, including marine, high humidity, rain and desert conditions (T), freezing precipitation (O)

**Rationale:** Corrosion and wear protective coatings provide FWS protection and contribute to system reliability throughout worldwide operational environments.

(7) Attribute 7 All FWS variants shall have Protective Lens Covers with lanyards or other means of protecting lens elements (T=O)

**Rationale:** Protective lens covers with a lanyard shall be necessary for normal use, storage and transport.

(8) Attribute 8 All FWS variants shall have an Input/Output (I/O) port. This I/O port shall facilitate for external power, head mounted or remote viewing displays and can be used for maintenance (T=O).

**Rationale:** The sight shall have a Head Mounted Display (HMD) to allow the Soldier to fire the MG without having to obtain eye relief while firing from the spade grips. The FWS can use an external monitor or HMD. Additionally, the I/O port can facilitate use of external power while at a stationary post or mounted in a vehicle to reduce battery requirements, and also perform maintenance.

(9) Attribute 9 Over the full operating temperature range the FWS shall provide a recognizable image within 120 seconds after FWS power is applied and within 3 seconds after being switched from stand-by to full on (T=O).

**Rationale:** This capability enables the Soldier to employ the sight quickly without delay and allows for the sight to toggle in and out of Standby which in turn increases battery life and extends operational readiness.

(10) Attribute10 Weapon boresight retention. The FWS, when mated to any of the weapons specified, or re-mated to the same weapon shall retain boresight over a 96 hour mission as defined by the OMS/MP in all environmental conditions, weapons fire shock, temperature, temperature shock, vibration, shock, mount/dismount cycles, input voltage drift, power on/off cycles (T=O).

**Rationale:** Weapon bore sight retention is critical for increased lethality and Soldier confidence during a 96 hour mission. There isn't always the ability to re-zero or confirm zero while on a 96 hour, continuous operation. However, generally a Soldier will be required to remove and re-install the sight once they stand-down or return to the FOB. During mission prep prior to leaving for another 96 hour mission they can confirm zero or conduct boresight validation.

Table 6-4 Weapon boresight retention

| FWS Type | Bracket/Rail System<br>MIL-STD-1913 rail | Retention (mrad) |
|---|---|---|
| Individual | M16A2,A4 & M4:5.56mm Rifle & Carbine | 0.5 |
| | M249 SAW: 5.56mm Squad Automatic Weapon | 0.5 |
| | MK16 (SCAR Light) MK17 (SCAR Heavy) | 0.5 |
| Crew Served | M240B: 7.62mm Medium Machine Gun | 2.0 |
| | M2 HB: 0.50 Caliber Machine Gun | 2.0 |
| | MK19 AGL: Grenade Machine Gun (40mm) | 5.0 |
| | MK47 Striker Automatic Grenade Launcher | 5.0 |
| | M16 and M4 Series: 5.56mm Rifle & Carbine | 0.5 |
| Sniper | M107 or M82A2: 0.50 Caliber Sniper Rifle | 0.5 |
| | M14, M24 & M110: 7.62mm Sniper Rifles | 0.5 |

(11) Attribute11 The FWS shall have a remote control wired (T) wireless (O), and at a minimum this device shall control polarity (black hot/white hot), electronic zoom and calibration.

**Rationale:** The Soldier needs the ability to perform these functions without removing his hands from the weapon. The Soldier needs to be able to place this control where it is comfortable to the Soldier. The polarity control allows the Soldier to more clearly see and recognize targets and then engage rapidly. The standard FWS view allows the Soldier to scan their area faster and then switch to a zoomed view using the electronic zoom for precise engagements. Calibration is done often to ensure the sight is optimized when observing a sector, giving the Soldier the best resolution possible prior to making engagement decisions.

7. Family of Systems and System of System Synchronization.

a. The Joint Requirements Oversight Council (JROC) approved Soldier as a System Initial Capabilities Document provides the foundation for supporting the development of the Core, Ground, Air, Mounted, and other related Soldier capabilities development. SaaS must provide individual Soldiers with a high probability of countering, withstanding and/or surviving all natural and/or manmade hostile environments within the full spectrum of military operations. Soldiers require protection while on a platform/Forward Operating Base (FOB) (air or ground). The FWS is specifically part of the Family of Optics solution proposed in the Small Arms CBA. It is linked to and supports capabilities described in that document as well as the NETT WARRIOR SoS and TRADOC CNA documentation. These links are based on the FWS supporting role or key enabler to a specific FoS and SoS topic area or capability.

b. Supported ICDs and Related CDD/CPDs**.**

(1) Combat Identification (CID).

(a) Accurate Situational Awareness (SA) and Target Interdiction (TI) capabilities, used in conjunction with Doctrine, TTP, and Rules of Engagement (ROE) can provide the needed knowledge for ground and air platforms whose primary function is enemy interdiction. On the non-contiguous battlefield, SA is an increasingly important factor that requires materiel and non-materiel solutions/capabilities promulgated throughout the mounted and dismounted force. TI requires a family of interoperable and complementary materiel capabilities, supported by DOTMLPF solutions from the individual skill level through collective skill sets. Supporting technologies, like near-infrared (night vision goggles) and far-infrared (thermal) optics are key to implementation of these basic SA and TI capabilities.

(b) There exists a capability gap of inadequate density of SA systems fielded on vehicles, aviation, and dismounted Soldiers which prevents timely dissemination of data. The minimum value associated with this gap is 1 each system per dismounted element, (Squad, Team, Observation Post (OP), etc.

(2) The NETT WARRIOR capabilities begin with improvements over the Land Warrior (LW) Increment II and build upon the capabilities to meet the needs of all Soldiers who conduct ground close combat in the Future Force. Land Warrior program is recognized as the initial impetus. This effort includes determining the optimal distribution of operational capabilities across teams and squads to maximize small unit mission performance and integration of capabilities to increase functionality while reducing weight, space, power consumption and logistical footprint. Combining the functions of daylight optics, night vision sensors, range-finding and target location in one device contributes to this increase in function while reducing the number of systems required.

(3) TRADOC Capabilities Needs Assessment. The FWS satisfies the requirements for: Enhanced Target capability; ability to operate across the depth and breadth of the battlefield; Employ kinetic weapons and effectively incorporate non-kinetic capabilities; Ability to create the necessary effects under all conditions and environments; and Support mobility requirements across the range of mobility operations.

(4) Soldier as a System (SaaS) ICD. The SaaS ICD identifies the need to provide Soldiers with optics that improve Situational Awareness and their ability to detect, recognize and

accurately engage targets with the individual, crew served and Sniper weapons during day, night and limited visibility/light conditions.

(5) Engagement Skills Trainer II (EST II) CPD. EST II integration will be considered during development of the FWS Capabilities Production Document (CPD) and future milestone decisions.

Table 7-1: Supported ICDs and Related CDD/CPDs.

| **Capability** | **CPD Contribution** | **Related ICDs & CDDs** | **Related CPDs** | **Joint Capability Area** | |
|---|---|---|---|---|---|
| | | | | **Tier 1** | **Tier 2** |
| Battlespace Awareness | Provides the Warfighter with the capability to detect, recognize, and rapidly react to hostile fires. | Ground Soldier System CDD<br>SaaS ICD<br>FCS ORD<br>EST II CPD | | Tier 1: Battlespace Awareness | Tier 2: Collection |
| Ability to conduct Joint Urban Operations | Provides Warfighter with the capability to exploit short-lived opportunities, and to ensure ground Soldiers find, fix, and destroy the enemy. | Ground Soldier System CDD<br>SaaS ICD<br>FCS ORD<br>EST II CPD | | Tier 1: Force Application | Tier 2: Maneuver and Engagement |
| Battle Command/ Situational Awareness | Provides the Warfighter with increased situational awareness allowing for rapid execution of maneuver. | Ground Soldier System CDD<br>SaaS ICD<br>FCS ORD<br>EST II CPD | | Tier 1: Command and Control | Tier 2: Understand and Decide |
| Lethality Detect, identify, and Employ lethal effects | Provides the Warfighter with the capability to exploit short-lived opportunities, and to ensure ground Soldiers find, fix, and employ lethal effects in all operational environments. | Ground Soldier System<br>SaaS ICD<br>FCS ORD<br>EST II CPD | | Tier 1: Protection | Tier 2: Collection |

8. <u>Information Technology and National Security Systems (IT and NSS) Supportability</u> N/A

9. <u>Intelligence Supportability</u> N/A

10. <u>Electromagnetic Environmental Effects (E3) and Spectrum Supportability</u>

a. <u>Electromagnetic Environment</u> The FWS shall not cause interference with other electronic equipment, and shall not be affected when operated in close proximity to other equipment, or interfere with or degrade the performance of existing platform/vehicle and dismounted operator instrumentation, weapons, sensors, or communications subsystems operating within its range.

b. <u>Spectrum Supportability</u> Procurement or acquisition of a wireless, spectrum dependent device to meet the objective requirement would be conducted IAW DOD guidance (e.g., DODD 3222.3, DODD 4650.1, DODI 4630.8, DODD 5000.1 and DODI 5000.2) as well as applicable Military Department (MILDEP) publications where required.

An application for equipment frequency allocation (i.e., DD Form 1494) would be initiated during development of the CPD prior to MS B if required. The DD Form 1494, Application for Equipment Frequency Allocation, may be releasable for coordination purposes to those foreign countries (host nations) in which permanent deployment or lengthy temporary use is contemplated.

The program manager (PM) acknowledges that if the system is spectrum dependent, the required spectrum support is or will be available in those host nations determined by the PM or procurer for the equipment's intended use.

The PM will develop a plan to obtain appropriate equipment allocation guidance/status where required prior to MS B or MS C as outlined in DODD 4650.1 in order to progress to the next phase.

11. Technology Readiness Assessment

a. Estimated Technology Readiness Levels. The PM has identified potential Critical Technology Elements for the FWS program. These components are estimated at Technology Readiness Level (TRL) 6 or higher when assessed by the PM in accordance with the DOD Technology Readiness Assessment Deskbook.

(1) Potential Critical Technology Elements. The Potential Critical Technology Elements for FWS are primarily based on components used in other weapon sights and night vision goggle devices, providing the FWS with a mature technology base.

(2) Long Wave InfraRed (LWIR) Focal Plane Arrays (FPAs). Uncooled LWIR FPAs have been used in TWS systems for more than six years, and are assessed at a TRL9. The systems have been fielded by the US Army since 2006 and are mission proven, with lean manufacturing processes applied through ManTech programs in coordination with the Night Vision Electronic Sensors Directorate (NVESD). FWS may chose to utilize the most recent advances in FPA development of small pixel pitch (≤ 12 micron). Small pixel advancements are currently in development and are assessed at TRL 6, based on work completed through an Industrial Base Innovation Fund program at NVESD.

(3) Digital NIR Sensor. The digital NIR sensor component of the FWS can be based upon multiple base technologies, both utilizing Complementary Metal-Oxide Semiconductor (CMOS) concepts. One of these is an Electron Bombardment Active Pixel Sensor (EBAPS) that has been manufactured in very small quantities for a recent fielding of prototype systems through RDECOM Ground Soldier Systems Distributed Aperture Program. EBAPS and Micro-Channel Plate (MCP) CMOS sensors have been demonstrated in ENVG (D) prototypes and demonstrated in an operational environment. This places both technologies at a TRL7.

(4) Display Technologies. The FWS program can utilize either Organic Light Emitting Diode (OLED) or Active Matrix Liquid Crystal Display (AMLCD) type displays to provide the information to the Soldier. A militarized version of the Commercial Off-The-Shelf (COTS) AMLCD technology has been proven in the field as part of the TWS program since its initial fielding as a part of the LWTS in 2004, and has been used in additional TWS and NVG systems

since. This technology is assessed as TRL9. The OLED technology is a COTS item and has been part of the fielded Marine Corps Individual Night Sight Thermal (IWNS-T) system, and can therefore be assessed as a TRL9. For both technologies, large format (≥1280x1024) arrays are considered for FWS in additional to the standard displays demonstrated in fielded systems. A larger format display has been integrated into the 17 micron LTWS systems and is completing developmental test and evaluation, demonstrating TRL8.

(5) RTA. A technology demonstration of the RTA capability required for the FWSI was performed at the US Army Expeditionary Warrior Experiment Spiral E at Fort Benning, Georgia, from August to December 2008. Demonstrated in a relevant environment, this technology is assessed as TRL6.

b. Manufacturing Readiness Challenges**.** No significant manufacturing challenges exist for the potential FWS material solutions, estimated by the PM at MRLs of 5 or higher in accordance with the DOD Technology Readiness Assessment Deskbook. Other programs within the Army and Marine Corps have, at a minimum, manufactured the components in low quantities and shown effective in the field. Some potential critical technology elements, like the FPAs and Display Technologies, have been produced in high volume for full-rate production programs successfully. Through the FWS developmental program, prototype materials, tooling, test equipment, and personnel skills shall be demonstrated in a near production prototypes that will be tested in an operational environment. Manufacturing challenges and risks shall be monitored through the developmental phase, and addressed prior to a full-rate production decision.

12. Assets Required to Achieve Initial Operational Capability (IOC) The FWS shall be issued to Heavy (HBCT), Infantry (IBCT) Stryker (SBCT) Brigade Combat Teams, 3rd Armored Cavalry Regiment, FWSI 18 per platoon, FWSCS one per M240, M2 & MK19, and FWSS one per sniper team. The FWS shall be issued to the USA Special Forces Command (USASFC), Special Warfare Center (SWCS) and 75th Ranger Regiment; FWSI at a rate of one per Operational Detachment A and B team member (ODA/ODB), FWSCS one per M240, M2, MK19, and MK47 weapons system, FWSS one per sniper weapon system. The current variants of TWS in the BCT's and USASFC shall be cascaded to Functional and Multi Function BDE's in accordance with ARFORGEN cycles. This BOIP is to provide organic capability to mounted and dismounted forces that must acquire and provide target recognition and far target location in all climates and battlefield conditions during both limited visibility and night missions.

Table 12-1 BOIP for all 3 variants of the FWS

| | FWSI | FWSCS | FWSS |
|---|---|---|---|
| IBCT | 15066 | 11484 | 129 |
| SBCT | 3420 | 4879 | 21 |
| HBCT | 4968 | 4970 | 78 |
| 75th RGR | 3009 | 704 | 216 |
| USASFC | 9980 | 3273 | 3201 |
| **Totals** | **36443** | **25310** | **3645** |

*Note: This table does not include floats, schools or war-time reserve/pre-position stock.*

13. Schedule and IOC and Full Operational Capability (FOC) Definitions. FWS IOC is attained with the completion of issue and New Equipment training (NET) completion of a BCT.

a. The Army shall achieve IOC when sufficient FWS assets are in place for a BCT level unit to apply the capability to combat operations, with training and maintenance in place by 3QFY17.

b. The First Unit Equipped (FUE) demonstrates a combat ready unit 2QFY17

c. Fielded systems are logistically supportable.

d. Full Operational Capability (FOC). The objective date for FOC is FY25.

14. Other DOTMLPF & Policy Considerations.

a. Doctrine. Can the warfighting capability be accomplished by a change in Doctrine? No, the doctrinal employment of ground Soldiers and Sniper teams is unlikely to change. The improved capability, not doctrine shall enable them to Detect, Recognize and aid in Identification of targets. The FWS shall enable Warfighters to more easily acquire and more accurately engage targets with small arms with enhanced threat identification by maintaining the operator's peripheral vision, and ultimately enable lethal target incapacitation by improving detection, recognition, and targeting processes. Changes in doctrine shall not increase the Soldiers ability to see at night or limited visibility conditions.

b. Organization. Can the warfighting capability be accomplished by a change in Organization? No, the warfighting capability to improve far target location, accuracy of engagements and responsiveness to the squad leader and platoon leader would not be effective by a change in organization.

c. Training. Can the warfighting capability be accomplished by a change in Training? No, improved training with current optics would not improve the warfighting capability and may only result in incremental improvement and will not fulfill the recognized requirements gap. A change in training without the required equipment shall only increase awareness; it shall not provide the warfighting capability required to accomplish the Army’s target location and engagement missions.

(1) System Training Plan (STRAP) Summary. FWS training shall produce Soldiers who are proficient in operating and maintaining FWS, leaders who can effectively plan for and employ it in combat operations, and units that can execute and sustain FWS operations and training. New Equipment Training (NET) shall provide initial proficiency for individual operator and maintenance skills, as well as leader’s employment skills. Institution training on employment of the FWS shall be integrated into Infantry, Armor, Engineer, Artillery, Military Police, Transportation, and Chemical, and Professional Military Education courses. Distance Learning (DL) based instruction for operators, maintainers, and leaders shall supplement training. Unit or operational training shall be responsible for sustaining individual and organizational proficiencies after NET, relying on unit trained NCOs, the training support package(s) delivered during NET, and DL based Interactive Multimedia Instruction (IMI) from the institution.

(2) Individual, Unit, and Crew Training. Individual training falls into three functions: Operator, Maintenance, and Employment. Operator and maintenance training shall be conducted in 2 phases: classroom instruction covering characteristics, operations, operator maintenance, and employment of the FWS; and hands-on training during live fire exercises and qualification ranges. Employment and leader training shall cover characteristics and capabilities of the FWS; planning and employment considerations; maintenance and sustainment; and an overview of TTPs. The FWS shall be integrated into unit collective training at team through platoon level. Operational unit training shall be responsible for sustaining individual and collective proficiency in FWS on their assigned weapons.

(3) New Equipment Training (NET). The FWS shall be fielded to units under the Unit Set Fielding (USF) concept at home station. Units shall be fielded the FWS, all applicable Training Aids, Devices, Simulators, and Simulations (TADSS), and Training Support Package (TSP) during USF. NET focuses on three functions–operations and maintenance, employment of FWS, and conducting unit sustainment training. NET shall be consolidated at Brigade Combat Team level (or higher where the fielding plan and unit schedules permit). The NET Team shall be composed of contractors and shall use a train the trainer approach. NET shall leverage computer based training, Interactive Multimedia Instruction, and Soldiers shall train on the newly fielded systems, and TADSS.

(a) Test Training Support Package (TTSP). The Training Developer and the Training Development Division from USAIS coordinates all deliverables from the materiel developer and submits the TTSP to the US Army Test & Evaluation Command IAW timelines provided by TRADOC Regulation 350-70 paragraph II-6-4.

(b) NET TSP. The Project Manager (PM) in conjunction with MCoE, Soldier Requirements Division (SRD) and Directorate of Training and Doctrine (DOTD) shall develop a TSP to support NET. It shall be based on the TTSP, modified by lessons learned during Operational Testing (OT). The TSP shall meet content requirements established in TRADOC Regulation 350-70.

(c) Interactive Media Instruction (IMI) Products. The Project Manager in conjunction with the MCoE SRD and DOTD shall develop IMI modules which shall support individual training in the institutional, operational, and self-development domains. The modules shall provide standalone computer based training as well as web-based training over the Internet.

(4) Reach-back Training. All institutional courses shall be available in IMI as either Computer Based Training (CBT) in a standalone digital media format or as web-based training hosted on the Army Learning Management System. Courseware shall comply with the latest version of the Shareable Content Object Reference Model (SCORM).

(5) Unit and Institutional TADSS. Simulators should be evaluated and considered for development in support of FWS training. These TADSS shall primarily support the operational training domain.

- Integration into Close Combat Tactical Trainer (CCTT). The capability to host simulations of FWS operations within the CCTT's Dismounted Soldier and the Virtual Soldier Multifunction Workstation would allow not only individual operator training but also its integration into collective training within CCTT.

- PM SMS is responsible for evaluating the feasibility of integration for the EST 2000 and would assume responsibility for concurrency and required upgrades in existing EST 2000 systems if integration was determined to be feasible. The final determination for integration will be the responsibility of the MCoE.

- PM SMS is responsible for evaluating the compatibility with Instrumentable MILES (IMILES) for any required changes, modifications, concurrencies, upgrades, and integration into the Army's Tactical Engagement Simulation Systems. The final determination for integration will be the responsibility of the MCoE.

(6) Combat Training Center Instrumentation and Interface Requirements. PM SMS is responsible that the FWS is compatible and does not interfere with Instrumentable MILES (IMILES) used at the Maneuver CTCs. All Live, Virtual, and constructive capabilities of the FWS will be evaluated for integration into the Mission Command Training Program and Combat Training Center (CTC) instrumentation. With regards to Live Force-on Force (FOF) training at the Maneuver CTCs, the PM will evaluate the resources required to provide the CTCs the means to validate the ability of units to employ within the force and mission rehearsal needs. The final determination for integration will be the responsibility of the MCoE.

(7) MOS Specific Training in the Institutional Training Base. There shall be no MOS specific training conducted at the institutional training base and this shall be validated during Pre-LUT Logistics Demonstration.

(8) Post Fielding Training Effectiveness Analysis (PFTEA). A post-fielding training evaluation ensures FWS training capabilities trains Soldiers, leaders, and units to standard. The PM shall fund a Maneuver Center of Excellence (MCoE) conducted PFTEA approximately 1-year following FUE.

d. Material. Can the warfighting capability be accomplished by a change in Material? Yes, the legacy TWS systems are heavy, use excessive battery power and don't provide the extended ranges required for operational success. Current equipment does not meet the Army or Joint requirements as outlined in the Small Arms CBA. The FWS is a fused weapon mounted optic and sensor system that provides thermal and low light image target acquisition under any light and environmental condition for the individual Soldier, Crew Served and Sniper weapon systems while dismounted or mounted. Additionally, it can be used for scanning and command & control by individuals in the hand held mode, or in-line with a day view optic without having to remove the day sight therefore allowing the same aiming TTPs and eliminating the need for re-zero. The RTA capability enables rapid offensive targeting, by reducing target acquisition and engagement timelines. Additionally, the RTA eliminates the active use of aiming lasers which in the past has put Soldiers at risk of revealing their position.

e. Leadership & Education. Can the warfighting capability be accomplished by a change in Leadership and Education? No, while improving leadership and education opportunities on current target detection, recognition, and identification processes would be a tremendous improvement in Soldier skills, it would not provide the Warfighter with increased night engagement ranges.

f. Personnel. Can the warfighting capability be accomplished by a change in Personnel? No, the quantity and quality of personnel does impact the warfighting capability. However, without the proper equipment and a good match between operators and maintainers, real situational awareness and battle space effectiveness won't occur.

g. Facilities. Can the warfighting capability be accomplished by a change in Facilities? No, a change in facilities alone cannot fulfill the required warfighting capability we must achieve. Constructive and Virtual simulations are reducing our reliance on live fire training as the only method of achieving proficiency in skills related to target detection, recognition and identification, however, these simulation scenarios are all based upon the employment of current inadequate night optics.

h. Logistics.

(1) Facility, Shelter, and Supporting Infrastructure. No new facilities are anticipated for maintenance of the FWS. Indoor secure storage shall be required as the FWS shall be considered a highly pilfered item IAW AR 190-51, Chapter 3, Para 3-6, AR 710-2, AR 735-2, DA Pam 710-2-1 and DOD 5100.76-M Physical Security of Sensitive Items. Organic small unit vehicles, MOLLE in possession of operators, standard shelters and maintenance vans assigned to units shall be used to store and support the FWS when in a field environment. If facilities are to be modified, then a Support Facility Annex shall be published by the ILS manager and included as an Appendix to this document.

(2) Transportability and Deployability. The FWS shall be transportable on standard road, rail, marine, or air transport systems. It shall also be capable of being deployed through air assault and airborne operations. The FWS shall be packaged to support transportation in a commercially acceptable container that shall adequately protect it from damage and the elements.

(3) Maintenance.

(a) Maintenance/Support Concept. The FWS leverages lessons learned from the predecessor TWS program and may continue to utilize the maintenance/support concept implemented (in-place) for that system. A Performance Based Logistics (PBL) will be considered and guide the decision-making regarding implementation to potentially produce better outcomes and reduced cost through increased reliability. Commensurate with the Maintenance Allocation Chart (Appendix B) from Technical Manual (TM) 11-5855-312-23&P, Technical Manual, Unit Maintenance and Direct Support Maintenance Manual (Including Repair Parts and Special Tools List), Sight, Thermal, 15 February 2005, maintenance tasks associated with field level repair under the Army Two Level Maintenance (TLM) System are limited to the replacement of easily accessible/replaceable components such as knobs, objective lens cover, eyecup, mount, carrying case, dust caps, reticle switch guard, battery door, battery stop, and transit/storage case. Under the Army maintenance system, repair beyond these parts replacement actions must be performed by the next higher level (sustainment), as substantiated by a "Level of Repair Analysis" (LORA). Unserviceable/Inoperable sight units must be returned to the IEW/RSC for direct-exchange with a serviceable unit. Procedures associated with the direct-exchange process are available to Army units in the form of a TWS Product Support Bulletin available from Product Manager Soldier Maneuver Sensors Logistics Manager for the AN/PAS-13 TWS.

(b) Maintenance Manpower Support. Requirements pertaining to maintenance manpower support are intended to address (constrain) the equipment maintenance burden placed on maintainers at the field level of repair. The manpower burden associated with field level maintenance performed in support of the FWS (as provided by dedicated field level Table of Organization and Equipment (TOE) maintenance personnel and associated contractor maintainers of the applicable Military Occupational Specialty (MOS)) shall not exceed 1.25 direct production man-hours over the course of a year, equating to a field level maintenance ratio of 0.00044 direct production maintenance man-hours (DPMMHs) per hour of system operation. Requirement precludes the FWS from needing maintenance manpower support in excess of that allocated to the currently fielded sight units the FWS shall replace/displace. The direct production maintenance man-hour support cited correlates directly to the man-hour expenditures allocated/documented in the Army Manpower Authorization Requirements Criteria (MARC) Maintenance Data Base (AMMDB) for repair of the fielded TWS (1.25 direct production field level man-hours expended annually), expressed in relation to the FWS's projected annual usage rate of 2,853 operating hours per year during wartime operations; this ratio equates to 0.00044 DPMMHs per operating hour, which enables the FWS's allowable maintenance burden to be determined for any test scenario based on the quantity of operating hours accrued by the FWS during testing. Field level maintenance shall not exceed 0.1 hours (as the maximum time to repair) for 95 percent of all maintenance functions performed by operators and maintainers (inspection/service/repair), commensurate with maintenance function work times defined in the Maintenance Allocation Chart for the predecessor TWS (as documented in TM 11-5855-312-23&P, Technical Manual, Unit Maintenance and Direct Support Maintenance Manual) (Including Repair Parts and Special Tools List); requirement precludes the FWS from increasing Army field level maintenance times above those defined for the predecessor systems it will replace/displace.

(c) Support Equipment. It is highly desirable that no new Test, Measurements and Diagnostic Equipment (TMDE) or Associated Support Items of Equipment (ASIOE) be required for the FWS. TMDE and ASIOE requirements shall be through the Logistics Demonstration conducted in accordance with the established maintenance concept. If required, new TMDE or ASIOE (compatible at field level with existing TMDE) shall be funded, developed and provided under the FWS program. All calibration requirements, procedures, and schedules will be identified in maintenance TM.

(d) Environmental Compliance Requirements. The user shall have the ability to train, operate, maintain, and dispose of the FWS system in full compliance with existing and known near term future environmental requirements contained in all applicable local, federal, and host nation statutes, standards, regulations and executive orders. Ozone depleting substance shall not be utilized in the production of the FWS.

(e) Technical Manuals (TM). TMs for Operators and Maintainers shall be produced to MIL STD and undergo a contractor validation and Government verification process to ensure accuracy and completeness. Electronic Technical Manuals (ETM) and Interactive Electronic Technical Manuals (IETM) shall be programmed for production. COTS digital manuals for operators and maintainers (used prior to the issuance of validated and verified MIL STD TMs), shall undergo a Government verification review prior to issue. Operator, Field and sustainment maintenance tasks shall be called out in the Maintenance Allocation Chart (MAC) found in the Field and Sustainment Maintenance Technical Manuals (TMs).

(f) Technical Data Package: The technical data package for each system variant shall be procured by the Government to accommodate cost effective material change, configuration control, re-procurement, and parts commonality requirements for organic support.

(g) Item Unique Identification (IUID). IUID is a DOD initiative that will enable easy access to information about DOD possessions that will make acquisition, repair, inventory, and deployment of items faster and more efficient. The implementation of IUID requirements means that qualifying items must be marked with a Unique Item Identifier (UII) in accordance with the DOD Guide to Uniquely Identifying Items. Specifically, MIL STD 130 requires that all FWS qualifying components, to include legacy components that transition through organic depots, must be marked with a UII in the form of a machine readable 2D Data Matrix, the contents of which will be encoded in the syntax of ISO/IEC 15434 and the semantics of ISO/IEC 15418 or the Air Transport Association (ATA) Common Support Data Dictionary (CSDD). All 2D Data Matrix bar codes must meet the verification standards for mark quality as established in ISO 15415 and SAE AS9132.

15. Other System Attributes. .The FWS has no unique environmental requirements and shall be designed to operate under all environmental conditions (except "severe cold") without the requirement for special kits.

a. Storage Temperature. The FWS shall operate in Hot, Basic, and Cold weather conditions and regions of the world. Climatic conditions defined in AR 70-38 for Hot, Basic, and Cold will be addressed in FWS development. The FWS must be capable of being operated and stored in the hot, basic, and cold climatic conditions design types. The FWS must be able to withstand the thermal shock associated with large and rapid changes in temperatures such as might be encountered during transfer to and from heated shelters in cold climates and from air conditioned structures to the ambient environment in hot climates.

b. Embedded Instrumentation. Built-in Test/Built-in Test Equipment (BIT/BITE) shall isolate and report failures or faults occurring in the system to the maximum extent possible. The extent of off-line BIT to isolate faults shall be determined by the LSA process.

c. Conventional Weapons Effects and Initial Nuclear Weapons Effects. The FWS is not mission critical. Initial Nuclear Weapon Effects of High-Altitude Electromagnetic Pulse (HEMP) survivability is not required.

d. Chemical, Biological, Radiological and Nuclear (CBRN) Survivability. The FWS shall operate and survive in CBR and nuclear environments with no performance impact. Decontamination for reuse is desired without materiel replacement of subcomponents. The FWS shall be operable and decontaminable by personnel in MOPP IV.

e. Expected Mission Capability. FWS shall be mission capable in all environments. The system must meet basic cold and hot weather conditions and remain operational in adverse weather conditions such as rain, fog, salt fog, snow, dust, and smoke.

f. Physical and Operational Security Needs. There are no anticipated new requirements for physical and operational security needs above current systems. The FWS shall be physically secured in the same manner as other sensitive items (i.e. Arms Room).

g. Human System Integration/MANPRINT.

(1) Manpower - Introduction of the FWS shall not increase the overall number of personnel, both, military and civilian, required to operate, maintain, and support the item.

(2) Personnel - The operation, maintenance, and support of the FWS shall not require aptitudes, skills, or capabilities beyond those currently present in the user population.

(3) Training - The instruction and resources required providing the Warfighter and maintainer with knowledge, skills and abilities in proper operation, maintenance, and support Army systems shall not increase due to the introduction of the FWS.

(4) Human Factors Engineering - The FWS design shall promote effective Soldier-machine integration for optimal total system performance. Design principles taking into account human capabilities and limitations shall be incorporated into system definition, design, development, and evaluation. This includes concepts of human-computer interface (e.g., ease of perception and comprehension of displays, ease of use of controls) and compatibility of FWS with other mission-essential equipment (including but not limited to use with standard combat gear, CBRN, and environmental clothing). The FWS should not interfere with the performance of common Soldier tasks. Equipment design must consider mission-dependent tasks and demands through consultation with SMEs, in order to maximize ease of use, minimize workload and enhance mission performance.

h. System Survivability.

(1) FWS will be hardened through Optical Augmentation (OA) signature reduction (Detection Avoidance) per the guidelines in the Army laser hardening standard, "U.S. Army Standards for Hardening O/EO Systems Against Lasers on the Battlefield". Detection Avoidance through reduction of the OA signature is a critical area of laser protection. (Objective)

(2) FWS will be hardened against laser jamming and damage of the sensors per the guidelines in the Army laser hardening standard, "U.S. Army Standards for Hardening O/EO

Systems Against Lasers on the Battlefield". The projected primary threat will come from Low-level Energy Lasers (LEL). (Objective)

(3) System Safety - The FWS design and operational characteristics shall minimize the possibilities for safety hazards to personnel and equipment.

(4) Health Hazards - The FWS design and operational characteristics shall not result in new or increased health hazards to personnel. A Health Hazard Assessment (HHA) will be requested from the Center for Health Promotion & Preventive Medicine (CHPPM) early in the development or procurement process. This HHA will be updated at each Milestone Decision Review (MDR) as required by AR 40-10.

(5) Soldier Survivability - The FWS shall have a positive effect on the overall survivability of the Soldier. It shall not create added vulnerabilities to users by creating signatures which the enemy could use to detect users. The FWSI provides a capability that allows Soldiers to rapidly engage targets without the need to change from goggles/aiming lasers to a thermal sight. The FWSCS provides a capability that allows for improved resolution and recognition at a distance greater than that of some existing systems by providing a fused sensors system and incorporates range to target and ballistic calculations to adjust the reticle. The FWSS is a clip-on (in-line) sight which allows the Sniper to use the thermal sight in conjunction with a day view optic.

i. Environmental Impact.

(1) The system design shall reduce or eliminate the use of hazardous chemicals or materials, energy and the generation of hazardous wastes during the manufacture, operation and maintenance of the FWS system. Ozone depleting substances shall not be utilized in the production of the FWS. The FWS system shall be designed to eliminate or minimize environmental impact IAW DoDI 5000.02

(2) The system shall be energy efficient, and minimize the use of external power sources as well as alternative and reusable energy sources, where possible.

16. <u>Program Affordability</u> The combat developer has been an integral part of the development team helping define and establish program goals and identifying where performance trades can be established while assuring that the operational needs of the Warfighter are addressed. All performance requirements have been defined in terms of capabilities at the system level with Key Performance Parameters (both threshold and objective values) established. All procurements shall evaluate the cost independently, as price is an independently evaluated component of source selections. As performance requirements are provided in a threshold and objective status, the value of obtaining objective values can be compared against the threshold - allowing for the value of capabilities above the threshold to be exchanged, substituted, or adjusted for the sake of another while at the same time ensuring cost targets are met.

a. Life-Cycle Costs: Project Manager Soldier Sensors and Lasers has developed an initial Life Cycle Cost Estimate (LCCE) for FWS using the Automated Cost Estimating Integrated Tools (ACEIT) system. The LCCE shall be a living document and updated as required as the

FWS Program evolves, to include an update in support of a Milestone B decision. The results shown in Table 16-1 were generated in 2QFY11. The estimates have been based on data from other programs that have been developed and fielded which employ like technologies/components, maintenance strategies, and procurement quantities. In addition, the Product Manager will apply Cost as Independent Variable (CAIV) principles to identify cost reduction opportunities in the Program.

Table 16-1 Life Cycle Cost (BY11$)

| **Objective** | **Threshold** |
|---|---|
| FWSI $23,020 | FWSI $25,320 |
| FWSCS $23,600 | FWSCS $25,960 |
| FWSS $23,150 | FWSS $25,460 |

b. FWS Affordability: Data from other programs fielding like technologies/components, maintenance strategies, and procurement quantities were evaluated and utilized in developing the support cost estimates for this program. The number of fielded systems by program year has estimated based on the BOIP developed for FWS and an estimated unit cost. Due to the nature of the sights being procured for the FWS, there are no Manpower or Military Construction costs associated with the FWS program. The Program Affordability for FWS is shown in Table 16-2 within the constraints of the budget funding allocated and was generated in 2QFY11. The cost for fielding the IOC is $375.624M, in TY$ FY11-17, and is 21% of the total program cost. The cost for fielding the FOC is $1627.373M, in TY$ FY11-24, and is 89% of the total program cost.

Table 16-2: Program Affordability of FWS FY11-17(TY $M)

<table>
<tr><th colspan="2"></th><th colspan="9">Program Affordability FY11-17 for Initially Fielded Units (IOC)</th></tr>
<tr><td colspan="2"></td><td>FY</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>Totals</td></tr>
<tr><td rowspan="9">Funding (TY $M)</td><td rowspan="3">RDTE</td><td>Required</td><td>14.275</td><td>10.979</td><td>8.654</td><td>5.202</td><td>5.083</td><td>5.291</td><td>5.299</td><td>54.783</td></tr>
<tr><td>Funded</td><td>14.275</td><td>10.979</td><td>8.654</td><td>5.202</td><td>5.083</td><td>5.291</td><td>5.299</td><td>54.783</td></tr>
<tr><td>UFR</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td rowspan="3">PROC</td><td>Required</td><td>0</td><td>0</td><td>0</td><td>0</td><td>43.659</td><td>134.671</td><td>141.043</td><td>319.373</td></tr>
<tr><td>Funded</td><td>0</td><td>0</td><td>0</td><td>0</td><td>43.659</td><td>134.671</td><td>141.043</td><td>319.373</td></tr>
<tr><td>UFR</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td rowspan="3">OMA</td><td>Required</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1.468</td><td>1.468</td></tr>
<tr><td>Funded</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1.468</td><td>1.468</td></tr>
<tr><td>UFR</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>

c. Cost Analysis: Project Manager Soldier Sensors and Lasers has begun development of a Program Office Estimate (POE) for FWS. This POE was independently validated by the RDECOM Cost Analysis Team in Aberdeen, MD on 3 Feb 2011. The estimates provided in

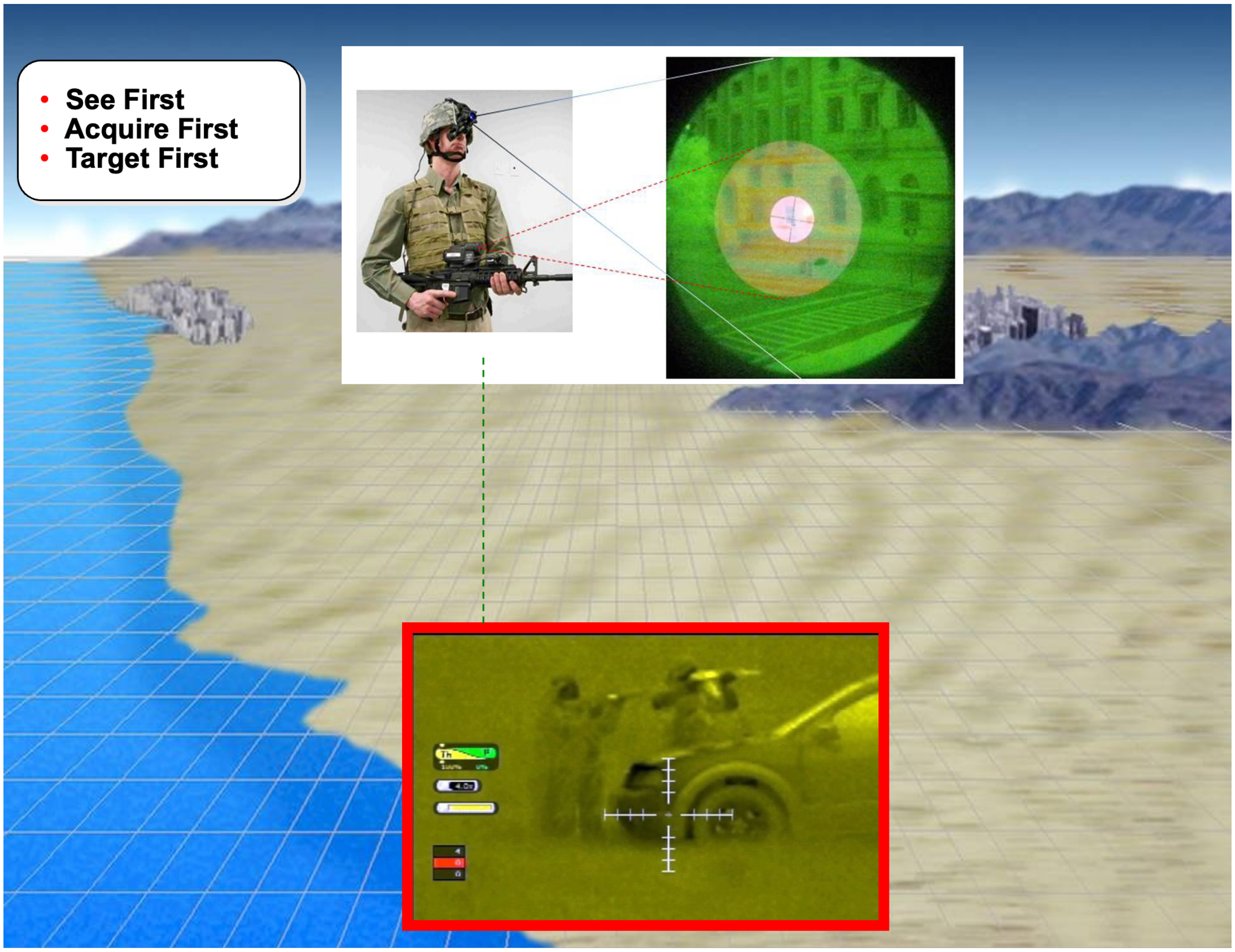
Table 16-3 are based on a PM calculation using estimated product costs and projected support estimates through FY45 for the OMA contribution. These estimates will be refined through the use of the ACEIT system and updated throughout the life of the program. In the Cost Analysis, the transition point for the FWS program from Increment I to II in RDT&E is FY15 and for Procurement is FY20. Both Increments are used to fulfill the total BOIP in Table 12-1.

Table 16-3: Cost Analysis – FWS

| **Cost Analysis (TY $M)** | | |
|---|---|---|
| Cost Heading | Increment 1 | Increment 2 |
| Total RDT&E | 39.109 | 21.213 |
| Total OPA | 737.214 | 785.581 |
| Total MPA | 0.0 | 0.0 |
| Total MCA | 0.0 | 0.0 |
| Total OMA | 111.693 | 131.118 |
| Total Program Requirement (TY) | 888.016 | 937.912 |

d. Source of funding. Funding for this program was transferred from the Thermal Weapons Sight Program.

Appendix A: Net Ready



Appendix B: References

Capstone Concept for Joint Operations (CCJO) Version 3.0, 15 Jan 09.

Chairman Joint Chief of Staff Instruction (CJCSI) 3170.01G, Joint Capabilities Integration and Development System (JCIDS), 1 Mar 09.

CJCSI 6212.01E, Interoperability and supportability of Information Technology & National Security Systems, 15 Dec 08.

Chairman Joint Chief of Staff Manual (CJCSM) 3500.04E, Universal Joint Task List (UJTL), 1 June 2011

CJCSI 3810.01C, Meteorological and Oceanographic Operations, 15 August 2009.

CJCSM 3170.01C, Operation of the JCIDS, 1 Mar 09.

DODD 3222.3, DOD Electromagnetic Environmental Effects (E3) Program, 8 Sep 04.

DOD Instruction 1322.26, Development, Management, and Delivery of Distributed Learning, 16 Jun 06

Army Regulation 71-9, Material Requirements, 28 Dec 09.

Army Regulation 70-1, Army Acquisition Policy, 31 Dec 03.

Army Regulation 70-38, Research Development, Test and Evaluation of material for Extreme Climatic Conditions, 15 Sep 79.

Core Soldier System CPD, JROC validated, 21 May 07.

FM 7-15, Army Universal Task List, Feb 09.

Ground Soldier System CDD, AROC validated, 1 Feb 06.

MIL-STD 882D DoD Standard Practice for System Safety

MIL-STD 1472F Human Factors Engineering

Mounted Soldier System CDD, Joint Certification, 5 Dec 06.

SaaS ICD, JROC validated, 21 Oct 05.

System Threat Assessment Report for the Future Combat System–Brigade Combat Team, dated 27 Feb 07.

TRADOC Pam 525-66, Military Operations: Force Operating Capabilities, 1 Jul 05.

TRADOC Pamphlet 525-66, Force Operating Capabilities, 01 Jul 05.

FWS Analysis of Alternatives, Nov 09.

Marine Crew Served Weapon Sight (MCSWS) Initial Capabilities Document (ICD) DTD: 1 Sep 2004

Infantry Marine Individual Weapon Sight (IMIWS) Initial Capabilities Document (ICD) DTD: 1 Sep 2004

Appendix C: Acronym List

Part I: Abbreviations and Acronyms

AAO – Army Acquisition Objective
ACAT- Acquisition Category
ACEIT - Automated Cost Estimating Integrated Tools
AGL- Automatic Grenade Launcher
ALDT- Administrative and Logistics Delay Time
AMLCD- Active Matrix Liquid Crystal Display
$A_o$ – Operational Availability
AOA- Analysis of Alternatives
ARCIC – Army Capabilities Integration Center
ARFORGEN- Army Force Generation
AROC – Army Requirements Oversight Council
ASIOE - Associated Support Items of Equipment
AT – Alert Time
ATR- Aided Target Recognition
AWST- Advanced Weapon Sight Technology
BBS - Brigade and Below Scenario
BCT – Brigade Combat Team
BDM- Bunker Defeat Munitions
BFV – Bradley Fighting Vehicle
BIT/BITE - Built-in Test/Built-in Test Equipment
BOIP- Bases of Issue Plan
CAIV- Cost as Independent Value
CBRNE – Chemical, Biological, Radiological, Nuclear and Explosive
CBT – Computer Based Training
CBA – Capability Based Assessment
CB-A- Cost Benefit Analysis
C3D2 - Cover, Concealment, Camouflage, Denial, Deception
CDD – Capability Development Document
CECOM – Communication Electronics Command
CHPPM - Center for Health Promotion and Preventive Medicine
CLS – Contractor Logistical Support
CMOS-Complementary Metal-Oxide Semiconductor
COTS – Commercial-Off-The-Shelf
CPD – Capability Production Document
CTC – Combat Training Center
CQB-Close Quarters
C4ISR – Command, Control, Communication, Computers, and Intelligence, Surveillance, and Reconnaissance
DARI- Detect, Acquire, Recognize and Identify
DEW – Directed Energy Weapons
DIA- Defense Intelligence Agency
DL – Distance Learning
DOD- Department of Defense

DOTD – Director of Training and Doctrine
DOTMLPF – Doctrine, Organization, Training, Material, Leadership and Education, Personnel, Facilities
DPMMH-Direct Production Maintenance Man-Hours
DT – Development Test
DVO-Day View Optic
E3 – Electromagnetic Environmental Effects
EA – Electronic attack
EBAPS-Electron Bombardment Active Pixel Sensor
EFF-Essential Function Failure
EMI – Electro-Magnetic Impulse
EMP – Electromagnetic Pulse
ENVG- Enhanced Night Vision Goggles
EO/IR-Electro-optics/Infrared
ETM-Electronic Technical Manual
FAA – Functional Area Analysis
FDSC - Failure Definition and Scoring Criteria
FCS – Future Combat Systems
FLIR – Forward Looking Infrared
FM- Field Manual
FNA – Functional Needs Analysis
FOB – Forward Operating Base
FOC – Full Operational Capabilities
FOF-Force on Force
FOS – Family of Systems
FOV – Field of View
FPA-Focal Plane Array
FRP – Full Rate Production
FSA – Functional Solution Analysis
FTX – Field Training Exercise
FUE – First Unit Equipped
FWS – Family of Weapons Sight
FWSI – Family of Weapons Sight Individual
FWSCS – Family of Weapons Sight Crew Served
FWSS – Family of Weapons Sight Sniper
HBCT – Heavy Brigade Combat Team
HEMP – High Altitude Electromagnetic Pulse
HHA-Health Hazard Assessment
HMD- Head Mounted Display
HMMWV- High Mobility Multi-purpose Wheeled Vehicle
I2 - Image Intensification
IAW-In Accordance With
IBCT – Infantry Brigade Combat Team
ICD – Initial Capabilities Document
ICS – Interim Contractor Support
IED – Improvised Explosive Device
IETM - Interactive Electronic Technical Manuals (IETM)

IEW/RSC- Intelligence and Electronic Warfare Regional Support Center
IMIWS-Infantry Marine Individual Weapon Sight
IMT – Individual Movement Techniques
IO – Information Operations
IOC - Initial Operational Capability
IR - Infrared
ISR – Identification Safety Range
IT – Information Technology
ITT – Integrated Concept Team
JCA – Joint Capability Area
JCF – Joint Contingency Force
JCIDS – Joint Capabilities Integration & Development System
JFC - Joint Future Capabilities
JOC – Joint Operational Capabilities
JOpsC - Joint Operations Concepts
JROC – Joint Requirements Oversight Council
KIP - Key Intelligence Position
KPP – Key Performance Parameter
LCC – Life Cycle Cost
LCCE – Life Cycle Cost Estimate
LED – Light Emitting Diode
LEL - Low-level Energy Laser
LFX – Live Fire Exercise
LIS - Logistics Information Systems
LLL- Low Light Level
LUT – Limited User Test
LW – Land Warrior
LWIR- Long Wave Infrared
METOC - Meteorological and Oceanographic
MILES – Multiple Integrated Laser Engagement System
Mil-Std- Military Standard
MCO – Major Combat Operations
MCoE- Maneuver Center of Excellence
MCSWS- Marine Crew Served Weapon Sight
MG- Machine Gun
MICH-Modular Integrated Communication Helmet
MOLLE - Modular Lightweight Load-bearing Equipment
MOA- Minute of Angle
MOPP – Mission Oriented Protective Posture
MOS- Military Occupational Specialty
MOUT – Military Operations in Urban Terrain
MRad – Miliradian
MSD – Maintenance Support Device
MTBEFF – Mean Time Between Essential Function Failure
MTBSA – Mean Time Between System Abort
MTL-Manufacturing Readiness Level
CBRN – Chemical, Biological, Radiological, and Nuclear

NBCC – Nuclear, Biological, and Chemical Contamination
NDI – Non-Developmental Items
NECS - Net-Centric Enterprise Services
NET – New Equipment Training
NIR- Near Infrared
NLT – No Later Than
NR-KPP – Net Ready-Key Performance Parameter
NSS – National Security System
NVESD- Night Vision Electronic Sensor Directorate
NVG – Night Vision Goggles
O-Objective
ODA- Operational Detachment Alpha
O&O – Operational and Organizational
OEF – Operation Enduring Freedom
O/EO - Optical and Electro-Optical
OIF – Operation Iraqi Freedom
OLED-Organic Light Emitting Diode
OMS/MP – Operational Mode Summary/Mission Profile
ONS – Operational Need Statement
ORD – Operational Requirement Document
OT – Operating Time
OT – Operational Test
PBL – Performance Based Logistics
PFTEA – Post Fielding Training Effectiveness Analysis
PM – Product Manager
PMCS – Preventive Maintenance Checks and Services
PME-Professional Military Experience
POI- Program of Instruction
ROE-Rules of Engagement
RTA-Rapid Target Acquisition
RSTA – Reconnaissance, Surveillance, and Target Acquisition
SA – Situational Awareness
SaaS – Soldier as a System
SACBA- Small Arms Capability Based Assessment
SASO-Stability and Support Operations
SBCT – Stryker Brigade Combat Team
SCORM – Shareable Content Object Reference Model
SDD – System Development Demonstration
SME – Subject Matter Expert
SOCOM – Special Operations Command
SOS – System of System
SRD- Soldier Requirements Division
SSC – Small Scale Contingencies
SSP - Simulation Support Plan
STAR- System Threat Assessment Report
STRAP – System Training Plan
STX – Situational Training Exercise

SU – Situational Understanding
SWCS-Special Warfare Center
SWIR-Short Wave Infrared
T- Threshold
TADSS – Training Aids, Devices, Simulators, and Simulations
TDH – Technology Demonstration Hardware
TEWT – Tactical Exercise without Troops
TI - Target Interdiction
TIS – Thermal Imaging System
TLM-Two Level Maintenance
TM - Technical Manuals
TMDE – Test, Measurements and Diagnostic Equipment
TRADOC – Training and Doctrine Command
TRL – Technology Readiness Level
TSP – Training Support Package
TTP – Tactics, Techniques, and Procedures
TTSP – Test Training Support Package
TWS – Thermal Weapon Sight
UA-Unit of Action
UE – User Experiment
UJTL – Universal Joint Task List
USAIC – United States Army Infantry Center
USASFC- United States Army Special Forces Command
USF – Unit Set Fielders
WARM – Wartime Reserve Modes

Part II: Terms and Definitions

Acquisition Category (ACAT): Categories established to facilitate decentralized decision-making and execution and compliance with statutorily imposed requirements. The categories determine the level of review, decision authority, and applicable procedures. The DOD 5000.2-R, part 1, provides the specific definition for each acquisition category (ACAT I through III).

Attribute: A quantitative or qualitative characteristic of an element or its actions.

Analysis of Alternatives (AoA): The evaluation of the performance, operational effectiveness, operational suitability, and estimated costs of alternative systems to meet a mission capability. The AoA assesses the advantages and disadvantages of alternatives being considered to satisfy capabilities, including the sensitivity of each alternative to possible changes in key assumptions or variables. The AoA is one of the key inputs to defining the system capabilities in the capability development document.

Capability: The ability to achieve a desired effect under specified standards and conditions through combinations of means and ways to perform a set of tasks. It is defined by an operational user and expressed in broad operational terms in the format of a joint capabilities Document, Initial Capabilities Document or a Joint Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel and Facilities (DOTMLPF) change recommendation. In the case of materiel proposals, the definition shall progressively evolve to DOTMLPF performance attributes identified in the Capability Development Document and the Capability Production Document.

Capability Production Document (CPD). A document that addresses the production elements specific to a single increment of an acquisition program.

Capability Development Document (CDD): A document that captures the information necessary to develop a proposed program(s), normally using an evolutionary acquisition strategy. The CDD outlines an affordable increment of militarily useful, logistically supportable, and technically mature capability. The CDD may define multiple increments if there is sufficient definition of the performance attributes (key performance parameters, key system attributes, and other attributes) to allow approval of multiple increments.

Capability Gaps: The inability to achieve a desired effect under specified standards and conditions through combinations of means and ways to perform a set of tasks. The gap may be the result of no existing capability, lack of proficiency or sufficiency in existing capability, or the need to recapitalize an existing capability.

Combat Identification: Combat Identification is the process of attaining an accurate characterization of detected objects in a Joint Battlespace to the extent that high confidence, timely application of military options and weapons resources can occur.

Concept of Operations (CONOPs): A verbal or graphic statement, in broad outline, of a commander's assumptions or intent in regard to an operation or series of operations. The CONOPs frequently is embodied in campaign plans and operation plans; in the latter case, particularly when the plans cover a series of connected operations to be carried out simultaneously or in succession. The concept is designed to give an overall picture of the operation. It is included primarily for additional clarity of purpose. Also called "commander's concept".

Detection: In Tactical Operations, the perception of an object of possible military interest but unconfirmed by recognition.

Environment: Air, water, land, living things, built infrastructure, cultural resources, and the interrelationships that exist among them.

Family of Systems (FoS): A set of systems that provide similar capabilities through different approaches to achieve similar or complementary effects. For instance, the Warfighter may need the capability to track moving targets. The FoS that provides this capability could include unmanned or manned aerial vehicles with appropriate sensors, a space-based sensor platform or a special operations capability. Each can provide the ability to track moving targets but with differing characteristics of persistence, accuracy, timeliness, etc.

Hugging: Hugging is the exploitation of civilians as a protective shield by "hugging" lethal, long-range artillery systems next to churches, schools and homes. The enemy believes that possible collateral damage and detrimental media coverage shall cause US commanders to be more hesitant in authorizing the use of indirect or aerial fires in populated areas.

Initial Capabilities Document (ICD): Documents the need for a materiel approach or an approach that is a combination of materiel and non-materiel to satisfy a specific capability gap(s). It defines the capability gap(s) in terms of the functional area, the relevant range of military operations, desired effects, time, and doctrine, organization, training, materiel, leadership and education, personnel and facilities (DOTMLPF) and policy implications and constraints. The ICD summarizes the results of the DOTMLPF analysis and the DOTMLPF approaches (materiel and non-materiel) that may deliver the required capability. The outcome of an ICD could be one or more joint DOTMLPF change recommendations or capability development documents.

Identification: In Ground Combat Operations, discrimination between recognizable objects as being friendly or enemy, or the name that belongs to the object as a member of a class.

Interoperability:

(1) The ability of systems, units, or forces to provide services to and accept services from other systems, units, or forces and to make use the services, units, or forces and to use the services so exchanged to enable them to operate effectively together.

(2) The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users. The degree of interoperability should be defined when referring to specific cases.

Joint Capability Area (JCA): JCAs are collections of similar capabilities logically grouped to support strategic investment decision-making, capability portfolio management, capability delegation, capability analysis gap, excess, and major trades), capabilities-based and operational planning. JCAs are intended to provide a common capabilities language for use across many related DOD activities and processes and are an integral part of the evolving CBP process.

Key Performance Parameters (KPPs): Those attributes or characteristics of a system that are considered critical or essential to the development of an effective military capability and those attributes that make a significant contribution to the key characteristics as defined in the Joint Operations Concept. KPPs are validated by the Joint Requirements Oversight Council (JROC) for JROC Interest documents, and by the DOD component for Joint Integration or Independent documents. Capability development and capability production document KPPs are included verbatim in the acquisition program baseline.

Key System Attribute (KSA): An attribute or characteristic considered crucial in support of achieving a balanced solution/approach to a key performance parameter (KPP) or some other key performance attribute deemed necessary by the sponsor. KSAs provide decision makers with an additional level of capability performance characteristics below the KPP level and require a sponsor 4-star, Defense Agency Commander, or Principal Staff Assistant to change.

Objective Value: The desired operational goal associated with a performance attribute, beyond which any gain in utility does not warrant additional expenditure. The objective value is an operationally significant increment above the threshold. An objective value may be the same as the threshold when an operationally significant increment above the threshold is not significant or useful.

Recognition: In Ground Combat Operations, the determination that an object is similar within a category of something already known; e.g., tank, truck, man.

Supportability: Supportability is a key component of system availability. It includes design, technical support data, and maintenance procedures to facilitate detection, isolation, and timely repair and/or replacement of system anomalies. This includes factors such as diagnostics, prognostics, real-time maintenance data collection, and human systems integration considerations.

Sustainability: The ability to maintain the necessary level and duration of operational activity to achieve military objectives. Sustainability is a function of providing for and maintaining those levels of ready forces, infrastructure assets, materiel, and consumables necessary to support military effort.

Sustainment: The provision of personnel, training, logistic, environment, safety and occupational health management, and other support required to maintain and prolong operations or combat until successful accomplishment or revision of the mission or of the national objective.

System of Systems (SoS): A set or arrangement of interdependent systems that are related or connected to provide a given capability. The loss of any part of the system shall significantly degrade the performance or capabilities of the whole. The development of a SoS solution shall involve trade space between the systems as well as within an individual system performance.

System Capabilities:  Measures of performance such as range, lethality, maneuverability, and survivability.

System Characteristics:  Design features such as weight, fuel capacity, and size.  Characteristics are usually traceable to capabilities (e.g., hardening characteristics are derived from a survival capability) and are frequently dictated by operational constraints (e.g., carrier compatibility) and/or the intended operational environment (e.g., CBRN).

Threshold:  A minimum acceptable operational value below which the utility of the system becomes questionable.

Appendix D: Analysis

**Section I**
Cost Benefit Analysis (C-BA).

**1. Problem Statement, Objective and Scope**

a. Problem Statement: Warfighters need to detect, recognize, and identify threats targets for engagement with organic or nonorganic fires at greater ranges with better resolution to reduce the possibility of fratricide while increasing lethality. Warfighters need the ability to employ a rapid targeting system during CQB or offensive missions. Warfighters need a sight that can range targets and provide an adjust aim point to accommodate crew served weapon systems that have longer range. Warfighters continue to need the ability to engage targets in complete darkness and limited visibility and also see current lasers. Additionally, Snipers do not want to remove their day view optics from their weapon. The current Thermal Weapon Sight (TWS) program will reach Full Operational Capability (FOC) in 1QFY17.

(1) The Army Capstone Concept (ACC) describes the broad capabilities the Army will require in the 2016-2028 timeframe. It describes how the Army will apply available resources to overcome adaptive enemies and accomplish challenging missions in complex operational environments. The evolving operational environment and emerging threats to national security will require continuous assessment of Army modernization. The ACC also establishes the foundation for subordinate concepts that will refine capabilities and identify others essential to ensuring Army combat effectiveness against the full spectrum of threats that the Army, as part of the joint force, is likely to confront in the future. This concept is the foundation for future force development and the base for subsequent developments of supporting concepts, concept capability plans, and the Joint Capabilities Integration and Development System (JCIDS) process.

(2) The ACC articulates how to think about future armed conflict within an uncertain and complex environment. It provides a foundation for a campaign of learning and analysis that will evaluate and refine the concept's major ideas and required capabilities. The aim of Army operations is to set conditions that achieve or facilitate the achievement of policy goals and objectives. Future enemies will constantly adapt and seek ways to overcome Army strengths and capitalize on what they perceive as our vulnerabilities. We operate where our enemies, indigenous populations, culture, politics, and religion intersect and where the fog and friction of war persists. The U.S. Army must maintain its core competency of conducting effective combined arms operations in close combat to employ defeat and stability mechanisms against a variety of threats. The U.S. Army must also hone its ability to integrate joint and interagency assets, develop the situation through action, and adjust rapidly to changing situations to achieve what this concept defines as *operational adaptability*[1].

(3) ARCIC identified "What We Need Army Forces To Do" at the Capabilities Integration Enterprise Forum in support of the ACC:

[1] Department of the Army, *TRADOC Pamphlet 525-3-0, The Army Capstone Concept, Operational Adaptability—Operating Under Conditions of Uncertainty and Complexity in an Era of Persistent Conflict,, 21 Dec 2009*

- Conduct combined arms maneuver; defeat the enemy in close combat and seize and retain the initiative
- Conduct area security operations over wide areas (including population security)
- Understand complex situations in width, depth, and context; achieve unity of effort with diverse partners
- Connect operations to strategy; ensure progress toward policy goals
- Conduct sustained engagement to build partner capacity, prevent conflict, and prepare for contingencies
- Overcome anti-access and area denial capabilities; help ensure freedom of action in maritime and aerospace domains
- Conduct reconnaissance in close contact with enemy and civilian populations
- Employ a combination of defeat and stability mechanisms to accomplish the mission
- Conduct and sustain operations from and across extended distances
- Protect information and communications systems; enable ability to fight degraded
- Operate under conditions of transparency (24-hour news, internet)

b. Objective: The fielding of the FWS shall increase lethality and improve situational awareness by extending the range a Warfighter can recognize and engage potential targets. The FWS, used in conjunction with small arms, shall ensure that commanders in the field have constant rapid lethal firepower. The FWS shall have a reduced Size, Weight and Power (SWaP) from the legacy TWS systems, while increasing SA with multi spectral content and ques. The Rapid Target Acquisition (RTA) shall allow Warfighters to engage targets reflexively without shouldering the weapon and where conditions prohibit the use of NIR laser pointers. By applying embedded ballistic equations to the reticle of the crew served weapon systems an increased lethality is achieved at longer ranges and under more stressing environmental conditions.

c. Scope: The FWS CDD uses an incremental approach with new capabilities that address gaps identified in the Small Arms Capability Based Analysis (SA CBA) and the Soldier as a System (SaaS) ICD.

**2. Assumptions and Constraints**

a. Assumptions:

1. Thermal weapon sights program will sunset and need a follow-on program to continue to provide improved technology to the Soldier.
2. The FWS program will enable our forces to stay ahead of our advisories and continue to “Own the Night”.
3. Capability gaps identified in the Small Arms Capabilities Based Assessment (SA CBA) will not fundamentally change in the current operational environment.
4. Increment I potential Critical Technology Elements are no less than Technology Readiness Level (TRL) 6.

b. Constraints:

1. Current POM and budget estimates show procurement of FWS systems through 2024 based on the BOI with 2 increments.

2. Target identification is a process and cognitive task—it is not simply based on what is observed but also includes consideration of the operational context, situational awareness, and applicable rules of engagement.

**3. Current state (the Status Quo).** The current state of the Thermal Weapon Sight (TWS) Program began from an Operational Requirements Document (ORD) dated 1998 and began fielding in 2000. TWS is an ACAT II program fully funded. There are multiple TWS manufacturers with different alpha designators. The currently fielded systems are the AN/PAS – 13C, D, & E and have three variants each as described below:

**Light Weapon Thermal System (LWTS) Base System** is the AN/PAS – 13C,D,E V(1) Thermal Weapon Sight. It has a 25 mm lens, 320x240 25 µm Focal Plane Array (FPA) providing an 18º Horizontal Field of View (HFOV), at a 60 Hertz (Hz) frame rate. The display has a 7.68 by 5.76 mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width. The display viewing distance is 18.8 mm. The system has 1.31 Magnification.

**Medium Weapon Thermal System (MWTS) Base System** is the AN/PAS – 13C, D, E V(2) Thermal Weapon Sight. It has a 51.5 mm lens, 640x480 25 µm Focal Plane Array (FPA) providing an 18º Horizontal Field of View (HFOV), at a 30 Hertz (Hz) frame rate. The display has a 7.68 by 5.76 mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width and viewing distance of 18.8 mm. The system has 1.33 Magnification.

**Heavy Weapon Thermal System – Sniper (HWTS) Base System** is the AN/PAS – 13C, D, E V(3) Thermal Weapon Sight. It has a 99 mm lens, 640x480 25 µm Focal Plane Array (FPA) providing a 9 º Horizontal Field of View (HFOV), at a 30 Hertz (Hz) frame rate. The display has a 7.68 by 5.77mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width and a viewing distance of 18.8 mm. The system has 2.55 Magnification.

**4. Alternatives with Cost Estimates.** Three viable alternatives exist: (1) sustain TWS (Status Quo), (2) provide Soldiers significantly increased capabilities with FWS in the Long Wave Infared (LWIR) Spectrum or (3) provide Soldiers significantly increased capabilities with FWS in the Medium Wave Infared (MWIR) Spectrum.

**a. Alternative 1: TWS Maintain the Status Quo**.

Continue procuring to the end of AAO and sustain the current systems from the 1998 ORD as stated in Status Quo above. Don’t take advantage of all technology improvements and allow the TWS program to achieve FOC without follow on research & development and subsequent production of the next generation of weapons sights. This would put our Soldiers behind the technology curve and possible send them to future conflicts with less than the best equipment available.

1. Quantifiable benefits: AN/PAS-13C, D, E is a significant enhancement over the legacy TWS systems (AN/PAS-13B). These include increased range capabilities on the LWTS

from 620m to 820m, MWTS from 1200m to 1430m, and HWTS from 2200m to 2753m. The weight is also reduced on the LWTS from 2.65lbs to 1.95lbs, MWTS from 5.0lbs to 3.1lbs and the HWTS from 5.3lbs to 3.9lbs. (FOUO)

2. Non-Quantifiable Benefits. Primarily by reducing Size, Weight and Power (SWaP) and improving resolution the overall acceptance and use of the TWS have improved and the Soldiers ability to engage targets at greater ranges and sustain mission readiness status over longer periods of time has continued to be a "good news" story in the Thermal Weapon Sight Program.

3 Cost of Alternative 1

(a) TWS Total Life Cycle Costs are depicted below and represent the costs for all alpha designators within the program. These costs are outlined in the TWS Acquisition Program Baseline (APB) approved 4 November 2010. RDT&E obligations were completed in FY10 and Procurement completes in FY15. Sustainment costs from the inception of the program until FY35 are projected to be approximately $984.9M (TY $M) for all TWS designators and variants.

| **Alternative 1** | |
|---|---|
| Then Year APB Objectives | Cost |
| Tot RDTE (FY91 - FY10) | $ 72.7M |
| Tot Procurement (FY95-FY15) | $ 2931.7M |
| Total O&M | $ 984.9M |
| **Program lifecycle cost TY $M** | **$ 3989.3M** |

**Table 4-1 Alternative 1 Life Cycle Costs**

(b) The direct cost of maintaining the Status Quo from the present time (FY11) through end of life in TY $M is $640.4M (Procurement) and $892.8M (O&M), $1533.2M total.

(c) Indirect costs of maintaining the Status Quo are:

- Maintains Current Capabilities – Does not leverage the improvements in sensor technology for improved recognition range or battery life that occur through new system development. Would halt technology available to Soldier, possibly allowing the enemy to catch up to or exceed our night vision capability.

- Limitations of Current Systems - Not providing the Soldier with tools to increase lethality like the FWS rapid target acquisition and ballistic solutions. No decrease in SWaP to lighten the Soldier's load.

**b. Alternative 2: FWS in the LWIR Spectrum.**

Even with the improvements from legacy TWS to current TWS outlined above in Alternative 1 there is still room for improvements. Leverage the Night Vision Lab's development of improved displays, uncooled LWIR small pixel focal plane arrays, and digital image intensification sensors on the component level parts for FWS. Use the Advanced Weapons Sight Technology (AWST) program to integrate the advanced technologies and demonstrate an advanced capability to the Warfighter. Document and approve the FWS CDD which will fund RDT&E to improve the FWS and close Soldier gaps as outlined in the SA CBA and the SaaS ICD.

1. Quantifiable Benefits: The size, weight and range improvements of the technology components will allow us to combine MWTS and HWTS into one FWS Crew Served (FWSCS) sight. This reduces the number of spare parts required at the regional support centers to maintain an additional variant saving a recurring cost of $8.6K per repaired system (cost for additional thermal imaging module, housing and telescope). Also saved are the costs to obtain NSNs for these items, write and validate the additional variant information in the technical manuals, and develop variant specific training materials. Decreased system power usage extends battery life and reduces the number of batteries required from 6 on the current HWTS to 4 for the FWSCS. This would cut initial fielding costs by 4 batteries per system saving $146K, and the recurring savings to the Soldier's unit would be $1,505K over 500 mission hours with the quantity of FWSCS systems planned to purchase to fulfill the Basis Of Issue Plan (BOIP) in the CDD.

2. Non-Quantifiable Benefits: The LWIR system is a lower power and lighter than the MWIR. SWaP reductions reduce Soldier burdens and increase battery life. Improved resolution increases recognition ranges which improves Soldier survivability and aids in reducing fratricide. Sending the weapon sight image and reticle to the goggles or Head Mounted Display (HMD) allows for an offensive capability. Introduction of ranging to target and automatic reticle adjustments allows for greater accuracy with our crew served weapons systems. Provides a clip-on technology to the sniper teams allows for the use of a thermal night sight without removing the zeroed day optic.

3. Cost of Alternative 2

(a) The FWS Program Office Estimate (POE) for the LWIR system was independently validated by the RDECOM Cost Analysis Team in Aberdeen, MD on 3 Feb 2011. The Cost Analysis shown below is derived directly from this POE and is shown in TY $M. The estimate provided is based on a PM calculation using estimated product costs and projected support estimates through FY45 for the OMA contribution. In the Cost Analysis, the transition point for the FWS program from Increment I to II in RDT&E is FY15 and for Procurement is FY20. Both Increments are used to fulfill the total FWS BOIP.

| Cost Analysis (TY $M) | | |
|---|---|---|
| Cost Heading | Increment 1 | Increment 2 |
| Total RDT&E | 39.109 | 21.213 |
| Total OPA | 737.214 | 785.581 |
| Total MPA | 0.0 | 0.0 |
| Total MCA | 0.0 | 0.0 |
| Total OMA | 111.693 | 131.118 |
| Total Program Requirement | 888.016 | 937.912 |

**Table 4-2 Alternative 2 Life Cycle Costs**

| | Program Affordability FY11-17 for Initially Fielded Units (IOC) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | FY | 11 | 12 | 13 | 14 | 15 | 16 | 17 | Totals |
| Funding (TY $M) | RDTE | Required | 14.275 | 10.979 | 8.654 | 5.202 | 5.083 | 5.291 | 5.299 | 54.783 |
| | | Funded | 14.275 | 10.979 | 8.654 | 5.202 | 5.083 | 5.291 | 5.299 | 54.783 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | PROC | Required | 0 | 0 | 0 | 0 | 43.659 | 134.671 | 141.043 | 319.373 |
| | | Funded | 0 | 0 | 0 | 0 | 43.659 | 134.671 | 141.043 | 319.373 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | OMA | Required | 0 | 0 | 0 | 0 | 0 | 0 | 1.468 | 1.468 |
| | | Funded | 0 | 0 | 0 | 0 | 0 | 0 | 1.468 | 1.468 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Table 4-3 Alternative 2 Funding Requirements FY11-17**

**c. Alternative 3: FWS in the MWIR Spectrum.**

1. Quantifiable Benefits: The range improvements of the technology components will allow us to combine MWTS and HWTS into one FWS Crew Served (FWSCS) sight. This reduces the number of spare parts required at the regional support centers to maintain an additional variant saving a recurring cost of $8.6K per repaired system (cost for additional thermal imaging module, housing and telescope). Also saved are the costs to obtain NSNs for these items, write and validate the additional variant information in the technical manuals, and develop variant specific training materials.

2. Non-Quantifiable Benefits: Improved resolution increases recognition ranges, which improves Soldier survivability and aids in reducing fratricide. Sending the weapon sight image and reticle to the goggles or Head Mounted Display (HMD) allows for an offensive capability. Introduction of ranging to target and automatic reticle adjustments allows for greater accuracy with our crew served weapons systems. Provides a clip-on technology to the sniper teams allows for the use of a thermal night sight without removing the zeroed day optic.

3. Cost of Alternative 3

(a) The FWS Program Office Estimate (POE) for the MWIR system was derived from the validated LWIR POE. The parameters modified in this estimate are the unit price ($40K in BY11 $) and the number of batteries required for system operation (8 AAs vs. 4 AAs for the LWIR system). The Cost Analysis shown below is derived directly from this MWIR POE and is shown in TY $M. The estimate provided is based on a PM calculation using estimated product costs and projected support estimates through FY46 for the OMA contribution. In the Cost Analysis, the transition point for the FWS program from Increment I to II in RDT&E is FY15 and for Procurement is FY20. Both Increments are used to fulfill the total FWS BOIP.

| **Cost Analysis ($M)** | | |
|---|---|---|
| Cost Heading | Increment 1 | Increment 2 |
| Total RDT&E | 39.109 | 21.213 |
| Total OPA | 1330.098 | 2466.811 |
| Total MPA | 0.0 | 0.0 |
| Total MCA | 0.0 | 0.0 |
| Total OMA | 90.083 | 167.296 |
| Total Program Requirement (TY) | 1459.290 | 2655.320 |

**Table 4-4 Alternative 3 Life Cycle Costs**

| **Program Affordability FY11-17 for Initially Fielded Units (IOC)** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | FY | 11 | 12 | 13 | 14 | 15 | 16 | 17 | Totals |
| Funding ($M) | RDTE | Required | 14.275 | 10.979 | 8.654 | 5.202 | 5.083 | 5.291 | 5.299 | 54.783 |
| | | Funded | 14.275 | 10.979 | 8.654 | 5.202 | 5.083 | 5.291 | 5.299 | 54.783 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | PROC | Required | 0 | 0 | 0 | 0 | 43.797 | 134.819 | 141.175 | 319.791 |
| | | Funded | 0 | 0 | 0 | 0 | 43.797 | 134.819 | 141.175 | 319.791 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | OMA | Required | 0 | 0 | 0 | 0 | 0 | 0 | 521.84 | 0521.84 |
| | | Funded | 0 | 0 | 0 | 0 | 0 | 0 | 521.84 | 521.84 |
| | | UFR | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Table 4-5 Alternative 3 Funding Requirements FY11-17**

**5. Alternative Selection Methodology and Criteria:** The Methodology used to evaluate the alternatives was to compare the Alternative 1 (TWS/Status Quo) against Alternative 2 (FWS LWIR) and Alternative 3 (FWS MWIR) and apply ratings based upon the capability meeting the Soldier's needs. The ratings provided by the respective Alternatives are (Full, Partial, or None). Factor criteria were SWaP, Operational Effectiveness, and Cost.

**6. Compare SWaP, Operational Effectiveness and Cost Alternatives:**

Alternative 1, for the critical SWaP factors the status quo TWS has done a great job provided improvements over the life of the program. However, the current TWS technology insertion program (AN/PAS-13F & G) has reduced the weight and unit price while increasing the range, but after the completion of this effort in FY11, there are no future P3I programs scheduled to further improve the SWaP.

Alternative 2, based on current form factors of working models the size fully meets the program weight thresholds. Current LWIR weapons sights consume roughly 2W of power in steady state, meeting or exceeding the power thresholds. Design changes through the development program of all three variants could have some increase in weight and power when adding the Laser range finder /ballistic computer and data processing for the RTA.

Alternative 3, reduced size and weight is a Key Performance Parameter and alternative 3, is at risk of an increase in system power usage and weight as a result of the cooler required for the MWIR sensor. Current estimates on cooled MWIR systems show 4W of power consumption in steady state, with additional power draw during initial cool down, which can take in excess of 2.5 minutes. This increase would result in reduced battery life and/or increases in the number of batteries required. This would negate quantifiable benefit savings in battery costs shown in alternative 2.. The increased number of batteries and addition of the sensor cooler would also adversely affect system weight.

| **Type** | **Size, Weight and Power (SWaP)** | | |
|---|---|---|---|
| **Solution** | **Size** | **System Weight** | **Power** |
| **Alt 1 Status Quo** | **N** | **N** | **P** |
| **Alt 2 FWS LW** | **P** | **F** | **F** |
| **Alt 3 FWS MW** | **F** | **P** | **N** |

**Table 6.1 Alternative Solution Comparison-SWaP**

b. Operational Effectiveness results. Table 6-2

Alternative 1, the status quo provides partial or no capability to meet FWS requirements. All six areas listed are Key Performance Parameter in the FWS CDD that cannot be met by this alternative.

Alternative 2, fully meets all six areas of operation effectiveness.

Alternative 3, can meet five of the six operational effectiveness requirements but would require in excess of 2.5 minutes of time for the sensor cool down.

| **Type** | **Operational Effectiveness** | |
|---|---|---|

| Solution | Range | Resolution | Clip-on | Rapid Targeting | Auto Adjusted Reticle | Start-up time |
|---|---|---|---|---|---|---|
| Alt 1 Status Quo | P | P | N | N | N | P |
| Alt 2 FWS LW | F | F | F | F | F | F |
| Alt 3 FWS MW | F | F | F | F F | N | |

**Table 6.2 Alternative Solution Comparison-Operational Effectiveness**

c. Cost results. Table 6-3

Alternative 1, the Status Quo is fully funded.

Alternative 2, at known cost the program is funded to approximately 21% of the anticipated Army Acquisition Objective, based on the BOIP, in the current FY12-17 POM, with procurement continuing through FY24 with extended POM projected funding. There are no POM UFRs for FWS. The total program cost in BY00$ is $1.245B according to the POE.

Alternative 3, estimates indicate that the unit price of the MWIR is significantly higher than the LWIR alternative and there are additional costs associated with the additional power draw of the system. The impact on the total program cost is an increase of $1.921B over alternative 2, with a total program cost in BY00$ of $3.166B.

| Factor | Cost |
|---|---|
| Alt 1 Status Quo | F |
| Alt 2 FWS LW | P |
| Alt 3 FWS MW | N |

**Table 6.3 Alternative Solution Comparison- Cost**

**7. Trade-Offs:**

a. Alternative 1 system is built “as is” and cannot be retrofitted or changed to improve in the SWaP.

b. Alternative 1 has no capability to provide an offensive capability that can be used while on patrol or moving through buildings and still maintain depth perception and situational awareness (SA) and rapid engagement capability. It is not a clip-on sight and doesn’t provide for Adjust Reticle.

c. Alternative 2 can provide SA capability via HMD or goggles, and further enhances current Close Quarters Battle (CQB) missions, also lends to reduced fratricide by increasing resolution with a fused image.

d. Operational Effectiveness. Alternative 1 provides a single channel thermal sight, but does not meet the Combat Developer's requirement to support image fusion for better resolution and SA. Alternative 1 does not meet the reduced size and weight requirement. Alternative 3 does not meet the weight and power requirement, however it can exceed the range requirement.

e. Alternative 2 can provide increased capability for both mission duration and offensive operations requirements.

f. Cost. Assuming 20 year life cycles for all Three Alternatives, in TY$M Alternative 1 will cost $3989M to complete fielding and sustainment and provide current capabilities through FY35. Alternative 2 is estimated to cost $1826M and would provide increased capabilities FY16-FY44. Alternative 3 is estimated to cost $ 4115M and would provide increased capabilities FY16-FY45.

**8. Bill Payers**:

FWS would take over the TWS funding line in FY15. If funded in accordance with the extended TWS funding profile beyond the current POM, the Alternative 2 would be funded to 100% AAO through FY24. There are currently no unfunded requirements for this alternative. Subject to funding decisions beyond the FY12 to 17 POM, either bill payers will be identified to support the FWS or adjustments to the FWS schedule or BOI can be adjusted to provide capabilities within existing funds. Risks associated with these tradeoffs are described below.

Risk Assessment: Risks to the stated benefits of FWS program are defined as cost, schedule, and technical performance.

a. Cost risk is failure to fully fund FWS requirements in the FY12-17 POM. Probability of occurrence is currently unknown, but an assignment of a Low risk is reasonable as no UFRs have been identified with Alternative 2. Failure to fund RDT&E efforts with the current budget will have a greater downstream impact on the materiel developer's ability to integrate key technologies. Mitigation strategies include either delay scheduled start of follow-on Increments (e.g., delay overall Program schedule) or push implementation of selected capabilities within an affected Increment downstream to the next scheduled Increment.

b. Schedule and Cost risks are directly related. If FWS cost requirements are not fully funded the probability of schedule slip is high, and the probability of Cost growth as a result of decreased volume pricing is high. Mitigation strategy to remain on schedule could be to deliver a reduced capability as planned and delay implementation of any unfunded capabilities to the next Increment.

c. Technical performance as a standalone risk is assessed as Low. Current Technology Readiness Levels for both FWS capabilities are adequate to implement the planned acquisition strategy *provided future funding requirements are met.* However, Technical Performance cannot be completely separated from Cost and Schedule. Key technologies that FWS will leverage rely upon military programs to maintain the market base and schedule delays could have an impact on the availability or capacity of system subcomponents. A moderate risk with a moderate probability of occurrence is placed on the technology manufacturing base when coupled with a potential schedule slip. Performance specifications and requiring best industry practices, open architectures, and obsolescence plans as contractual requirements are mitigation strategies.

**9. Results and Recommendations:**

a. Results of the analysis.

(1) Alternative 1 provides a lower unit cost on average and is the least expensive alternative exclusive of any benefit provided. The Status Quo provides limited capabilities that do not meet the needs of the Warfighter and beyond FY11 has no future initiatives planned to improve KPPs.

(2) Alternative 1 is fully funded in the current POM and will cost an estimated $3989M in TY dollars ($2932M Procurement and $985M O&M) through life cycle completion in FY35.

(3) Alternative 1does not provide adequate Ranges or Operational Effectiveness to meet Combat Developer requirements nor can it contribute as effectively as Alternative 2 in providing an offensive night sight for the Warfighter.

(4) The TWS program will reach FOC in 1QFY17 and the FWS CDD is designed to continue the development of smaller, lighter, better night sights that address the gaps identified in the Small Arms CBA.

(5) Alternative 3 can partially or fully meet the KPPs and Operational Effectiveness requirements for the FWS with the exception of power consumption. The increased unit and battery costs with this alternative more than double the total program life cycle costs of alternative 2.

b. Recommendation: Provide Alternative 2 to the Warfighter. While the cost of maintaining the Status Quo is less expensive on a per-unit basis, one must consider the reduction in combat power. The Alternative 2 capabilities provide a reduction in Soldier burden by reducing the size and weight, 20% overmatch of like enemy weapon systems and will provide increased offensive operational benefits – all of which are unattainable with the Status Quo. Alternative 2 can also provide these capabilities with significant cost savings over the alternate sensor technology identified in Alternative 3.

**<u>Section II</u>**

**Analysis of Alternatives (AoA)**

1. **1. Family of Weapons Sight**

The United States Army has developed image intensification and thermal weapon sight technologies to provide the Warfighter the ability to own the night. Initially, Image Intensifiers ($I^2$) devices satisfied the original requirements. Additional capability gaps were identified as $I^2$ devices were deployed in theater. The Thermal Weapon Sight, Operational Requirements Document (ORD) identified requirements to provide the need for combat forces to acquire and engage targets during day, night and adverse environmental conditions. [2]

Development of the Thermal Weapon Sight (TWS) followed with a desire to improve the performance, Size, Weight and Power (SWAP). The Wexford Study conducted an evaluation of new technologies to improve the performance and provide recommendations for product changes grounded in operational effectiveness to ensure successful implementation and acceptance by soldiers. Wexford determined that soldiers use the TWS as a handheld device instead of on the weapon as a weapon sight because it lacks modularity, and wider spectrum of capabilities such as direct view, day, and night, no power, and image intensification.[3]

Program Manager for Optics and Non-lethal Systems (PM ONS) funded the Family of Individual Optics (FOIO) Business Case Analysis (BCA) to review technological shortcomings of current and near-term optics systems primarily based on risk derived investment strategies.[4] The systems reviewed included but not limited to day optics, image intensifiers, thermal devices, handheld, and head mounted displays.

In addition, the U.S. Army Capabilities Integration Center (ARCIC) tasked the U.S. Army Infantry Center (USAIC) to complete a Joint Capabilities Integration and Development (JCIDS) Capabilities Based Assessment (CBA) for Small Arms (SA) weapons. The result is the Small Arms Capabilities Based Assessment (SA CBA) (2008) that identifies the capability gaps for small arms and ancillary devices such as thermal, image intensifiers, and day optics.[5]

The draft Fused Weapon Sight Technical Trade Study (FWS TTS) reviewed developmental programs to close the gaps identified by each of the documents above, most notably the SA CBA.

---

[2] The United States Army (1998), Operational Requirements Document (ORD) for the Thermal Weapon Sight (TWS), p. 1.

[3] Product Manager Soldier Sensors and Lasers (PM-SSL). (2007). *Preliminary Analysis and Recommendations for Thermal Weapon Sight II (TWS) Technology Direction and Product Improvements,* 20 December 2007, p. 8.

[4] Program Manager, Optics and Non-Lethal Weapon Systems Infantry Weapons Systems Directorate Marine Corps Systems Command (2008), *Family of Individual Optics (FOIO) Business Case Analysis (BCA).*

[5] The United States Army Infantry Center (USAIC), *Small Arms Capabilities Based Assessment,* 2 April 2008.

## 2. Description of Alternative

The Doctrine, Organization, Training, Leadership, Materiel, Personnel, Facilities (DOTLMPF), Statutes, Regulations and Grants (SRG), Mission Needs Statements, PM SSL comments, and USAIC comments guided the alternatives selections. A precursor effort (Fused Weapon Sight Technical Trade Study) provided the basis for the family of weapon sights approach and the specific alternatives considered in this report. The alternatives were approved by Product Manager Soldier Sensors and Lasers (PM-SSL) and the United States Infantry Center (USAIC).The base system and alternatives support the Individual Rifle, Light & Medium Machine Gun, Heavy Machine Gun, and Sniper Rifles. Thermal system variants from the Thermal Weapon Sight II (TWS II) program precede this study and form the baseline systems for all weapon's classes. The PM SSL TWS II Improvements Study (Dec 2007) [6] stated that thermal imaging is required by individual soldiers as well as a spectrum of capabilities; such as previously funded Image Intensifiers ($I^2$); therefore, $I^2$ devices are alternatives in this study. Recently funded versions of the thermal weapon sight as well as improved versions of both thermal and $I^2$ variants defined by the Study Team are also alternatives. Appendix H contains summary descriptions of key technologies identified in the Fused Weapon Sight Technical Trade Study (FWS TTS). Alternatives in this study include systems identified in the FWS TTS conclusions:

- Develop a thermal system incorporating 25µm 320 x 240 uncooled thermal technology; although there is a bias in newer systems to apply 640x480 and larger formats, the 320x240 format is adequate for some applications and represents a mature, low power alternative.
- Develop advanced optics with an emphasis on Adaptive Polymers and Annular Folded Optics based on comparative analysis of Increment III technologies that may provide a significant benefit [7]
- Develop Short Wave Infrared (SWIR) detectors for Heavy Machine Gun applications
- Develop alternatives based on OLED and digital fusion – with the assumption that investment will continue in these areas.

The alternatives were presented and deemed suitable by Program Manager Soldier Sensors and Lasers (PM-SSL) and the United States Army Infantry Center (USAIC) for the AoA. USAIC requested the HISS alternatives and dividing the Sniper Rifle weapon class to two sub classes with separate baseline alternatives providing no additional data was required.

### a. Individual Rifle Alternatives

**Base System** is the AN/PAS – 13D V(1) Thermal Weapon Sight. It has a 25 mm lens, 320x240 25 µm Focal Plane Array (FPA) providing an 18º Horizontal Field of View (HFOV), at a 60 Hertz (Hz) frame rate. The display has a 7.68 by 5.76 mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width. The display viewing distance is 18.8 mm. The system has 1.31 Magnification.

---

[6] Product Manager Soldier Sensors and Lasers (PM-SSL). (2007). *Preliminary Analysis and Recommendations for Thermal Weapon Sight II (TWS) Technology Direction and Product Improvements,* 20 December 2007, p. 8.

[7] Fused Weapon Sight Study Team (2009). *Fused weapon sight technical trade study final briefing* [Power Point Slides]. CACI Technologies, Inc; Slide 46.

**Alternative 1** is a legacy image intensifier designed with a 27 mm lens, and GEN III 18 mm tube providing 40 º circular FOV. The eyepiece focal length is 2.7 cm with 1x Magnification. **Alternative 2** is a clip on thermal sight on Low Rate Initial Production (LRIP) with a 48.2 mm lens, 320x240 25µm FPA providing 10º HFOV at 60Hz. It has a 9.6 by 7.2 mm LCD with a 15 µm spot height and width. The display viewing distance is 5.74 cm with 1x Magnification and Electronic Zoom (EZOOM) up to 2x.

**Alternative 3** is an Increment I clip on thermal sight with a 21 mm lens, 320x240 17µm FPA providing 14° HFOV at 60Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The display viewing distance is 3.92 cm. This alternative contains technologies that are readily available for production. The system has 1x Magnification.

**Alternative 4** is an Increment I image intensifier. It has a 74 mm lens, and GEN III 18 mm tube providing 14 º circular FOV. The eyepiece focal length is 7.4 cm. The technology is extremely mature and in production; this study shall determine the maximum performance for an image intensifier based on this study's requirements. The system has 1x Magnification.

**Alternative 5** is an Increment II optically fused thermal and $I^2$ clip – on sight. A similar configuration is on Low Rate Initial Production (LRIP). This configuration has a 39.5 mm lens, 320x240 17µm FPA providing 7º HFOV at 60Hz. The thermal channel has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 7.4 cm. The $I^2$ component is a 74 mm lens, and GEN III 18 mm tube providing 14 º circular FOV. The eyepiece focal length is 7.4 cm. Note that the thermal viewing distance and $I^2$ eyepiece focal length are equal. The individual modalities and overall system have 1x Magnification.

**Alternative 6, and 6 (b)** are Increment III digitally fused thermal and $I^2$ clip – on sights with a 21 mm lens, 320x240 17µm FPA providing 14º HFOV at 60Hz. **Alternative 6 (c)** differs with a 39.5 mm lens to provide an optimal range performance in the thermal channel with a 7° HFOV. The $I^2$ components have a 60.2 mm lens, 1280x1024 11µm Electron Bombarded Active Pixel Sensor (EBAPS) providing 13 º HFOV at 30 Hz. Digitally fused systems share one display because a processor fuses the images prior to imaging. The display is a 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The display viewing distance for both systems is 3.92 cm. The individual modalities and overall system have 1x Magnification. Alternative 6 (b) and 6 (c) are low power systems with simple color fusion techniques.

**Alternative 7** is an Increment III Short Wave Infrared (SWIR) clip on sight with a 65.3 mm lens, 640x512 25µm FPA providing 14° HFOV at 30Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a display viewing distance of 3.67 cm. The system has 1x Magnification. This alternative contains technologies that are Commercially-Off-The Shelf (COTS) but need to be ruggedized for military deployment.

**Alternative 8** is an Increment III thermal clip on similar to Alternative 3, but with annular folded optics. This system requires a 33 mm diameter lens as opposed to an 18mm lens to capture the same amount of light. Like Alternative 3, it has a 320x240 17µm FPA providing 14° HFOV at 60Hz. The display is a 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The display viewing distance is 3.92 cm.

**Alternative 9** is an Increment III digitally fused thermal and Digital Image Intensifier ($DI^2$) clip – on sight similar to Alternative 6 but with annular folded optics. This system requires a 33 mm thermal and 82 mm $I^2$ aperture diameter lens to capture the same amount of light used with a conventional lens. This alternative has a 320x240 17µm FPA providing 14º HFOV at 60Hz and $I^2$ 1280x1024 11µm EBAPS providing 13 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The display viewing distance is 3.92 cm. The individual modalities and overall system have 1x Magnification.

**Alternative 10** is an Increment III image intensifier similar to Alternative 4 but with adaptive polymer optics. It uses mature GEN III 18mm $I^2$ tube technology; however, this system uses less mature adaptive polymer optics to provide optical zoom from 1 to 2x. The system has a 74 mm lens and 14º circular FOV at 1x and 7 º circular FOV at 2x. The eyepiece focal length is varies from 7.4 cm to 4.0 cm.

**b. Light and Medium Machine Gun Alternatives**

The PM SSL TWS II Improvements Study (Dec 2007) recommended eliminating the medium variant TWS II claiming the light variant could be used for close quarter engagement operations and the heavy variant for surveillance.[8] After consultations with PM SSL and USAIC, this study combined Light and Medium Machine Guns into a single sight approach for both weapons. The alternatives include the light and the heavy TWS II variants as options in this weapon class to measure against this study's Measures of Performance (MOP) and Measures of Effectiveness (MOE) metrics; on this basis, alternatives 1 and 2 are unique to this weapon class.

**Base System** is the AN/PAS – 13D V(2) Thermal Weapon Sight. It has a 51.5 mm lens, 640x480 25 µm Focal Plane Array (FPA) providing an 18º Horizontal Field of View (HFOV), at a 30 Hertz (Hz) frame rate. The display has a 7.68 by 5.76 mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width and viewing distance of 18.8 mm. The system has 1.33 Magnification.

**Alternative 1** is the AN/PAS – 13D V(1) Thermal Weapon Sight. It has a 25 mm lens, 320x240 25 µm FPA that providing an 18º HFOV, at a 60 Hz. The display has a 7.68 by 5.76 mm LCD with a 12 µm display spot height and width with a display viewing distance of 18.8 mm. The system has 1.31 Magnification.

**Alternative 2** is the AN/PAS – 13D V(3) Thermal Weapon Sight. It has a 99 mm lens, 640x480 25 µm FPA providing a 9º HFOV, at a 30 Hz. The display has a 7.68 by 5.76 mm LCD with a 12 µm display spot height and width and display viewing distance of 18.8 mm. The system has 2.55 Magnification.

**Alternative 3** is a standalone thermal sight in a currently funded program with a 31.1 mm lens, 640x480 17µm FPA providing 20º HFOV at 30Hz. It has a 9.6 by 7.2 mm LCD with a 15 µm spot height and width and viewing distance of 2.35 cm. The system has 1.17 Magnification.

**Alternative 4** is an Increment I standalone thermal sight similar to Alternative 3 optimized to meet the requirements in this study. It has a 33.5 mm lens, 640x480 17µm FPA providing 19º

[8] Product Manager Soldier Sensors and Lasers (PM-SSL). (2007). *Preliminary Analysis and Recommendations for Thermal Weapon Sight II (TWS) Technology Direction and Product Improvements,* 20 December 2007, p. 22.

HFOV at 30Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 2.35 cm. The system has 1.27 Magnification. The technologies are readily available for production.

**Alternative 5** is an Increment II image intensifier with Electron Bombarded Active Pixel Sensor (EBAPS) technology. It has a 44.1 mm lens, 1280x1024 11µm Active Pixel Sensor (APS) providing 18 º HFOV at 30 Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and a viewing distance of 2.35 cm. The system has 1.22 Magnification.

**Alternative 6** is an Increment II optically fused thermal and Digital Image Intensifier ($DI^2$) standalone sight. It is a 33.5 mm lens, 640x480 17µm FPA providing 19º HFOV at 30Hz. The $I^2$ component is a 79.2 mm lens, 1280x1024 11µm EBAPS providing 10 º HFOV at 30 Hz. This configuration has a prism merge the two images from the thermal and $I^2$ displays to the eye. Each of the two displays are 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The $I^2$ display has a 4.05 cm viewing distance and the thermal display has a 2.35 cm viewing distance through the prism to limit parallax issues. The individual modalities and overall system have 1.27 Magnification. The optical fusion approach was originally developed to merge $I^2$ tube optical output with thermal images from a micro display. $I^2$ tubes directly project an optical image without the use of digital processing electronics. Developments in digital $I^2$ detectors have led to $I^2$ technologies that produce digital image outputs suitable for micro displays, and designers can now combine digital $I^2$ and thermal images electronically for a single display. Optical fusion is still an option for use with digital $I^2$ systems, and while it constrains selection of display optics, it does perform an optical registration and blending process that is complex and power intensive in digital processing. The optical fusion approach trades complexities in optical and mechanical design during the development process for system power and mission duration during operations. PM SSL guidance[9] was to focus on solutions that meet requirements, versus solutions that dovetail with longer-term development goals. Therefore, this report includes both optical fusion and digital fusion as alternatives.

**Alternative 7 and 7 (b)** are Increment III digitally fused thermal and $DI^2$ standalone sight with a 33.5 mm lens, 640x480 17µm FPA providing 19º HFOV at 30Hz. The $I^2$ component is a 46 mm lens, 1280x1024 11µm EBAPS providing 17 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 9.6 by 7.2 mm OLED with a 15 µm spot height and width. The display viewing distance for both systems is 2.35 cm. The individual modalities and overall system have 1.27 Magnification.
This study assumed advanced digital fusion algorithms for digitally fused systems. Alternative 7(b) is a low power alternative with simple color digital fusion with both $I^2$ and thermal channels running at 30 frames per second (fps) but with the same optical properties as Alternative 7.

**Alternative 8** is an Increment III thermal standalone similar to Alternative 4 but with annular folded optics. This system requires a 45 mm diameter lens as opposed to a 28mm lens to capture the same amount of light. Like Alternative 4, it has a 33.5 mm focal length, 640x480 17µm FPA providing a 19 º HFOV at 30Hz and a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a viewing distance of 2.35 cm. The system has 1.27 Magnification.

---

[9] Meeting Minutes [Word]. *Future Weapon Sight Technical Trade Study, Initial Meeting Minutes*; May 23, 2008.

**Alternative 9** is an Increment III standalone image intensifier with EBAPS similar to Alternative 5 but with annular folded optics technology. This system requires a 60 mm diameter lens as opposed to a 37mm lens to capture the same amount of light. Like Alternative 5, it has a 44.1 mm focal length, 1280x1024 11µm Active Pixel Sensor (APS) providing 18 º HFOV at 30 Hz and a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a viewing distance of 2.35 cm. The system has 1.22 Magnification

**Alternative 10** is an Increment III digitally fused thermal and $DI^2$ standalone similar to Alternative 7 but with annular folded optics. This system requires a 45mm diameter thermal and 63mm $I^2$ lens to capture the same amount of light. It has a 640x480 17µm FPA providing 19º HFOV at 30Hz and an $I^2$ 1280x1024 11µm EBAPS providing 17 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a viewing distance for both systems of 2.35 cm. The individual modalities and overall system have 1.27 Magnification.

**Alternative 11** is an Increment III image intensifier similar to Alternative 5 but with adaptive polymer optics. This system uses EBAPS technology with adaptive polymer optics to provide optical zoom from 1.22 to 3x. It has a 44.1 mm lens, 1280x1024 11µm providing an 18 º HFOV at 1.22x and when compressed to 109 mm effective focal length provides a 7 º HFOV with 3x magnification. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 2.35 cm.

**c. Heavy Machine Gun Alternatives**

**Base System** is the AN/PAS – 13D V(3) Thermal Weapon Sight. It has a 99 mm lens, 640x480 25 µm Focal Plane Array (FPA) providing a 9 º Horizontal Field of View (HFOV), at a 30 Hertz (Hz) frame rate. The display has a 7.68 by 5.77mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width and a viewing distance of 18.8 mm. The system has 2.55 Magnification.

**Alternative 1** is a standalone thermal sight in a currently funded program with a 65.1 mm lens, 640x480 17µm FPA providing a 10 º HFOV at 30Hz. It has a 9.6 by 7.2 mm LCD with a 15 µm spot height and width and viewing distance of 2.35 cm. The system has 2.45 Magnification.

**Alternative 2** is an Increment I standalone thermal sight similar to Alternative 1 optimized to meet the requirements in this study. It has a 126 mm lens, 640x480 17µm FPA providing 5 º HFOV at 30Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 2.35 cm. The system has 5x Magnification. The technologies are readily available for production.

**Alternative 3** is an Increment II image intensifier with Electron Bombarded Active Pixel Sensor (EBAPS) technology. It has a 127.1 mm lens, 1280x1024 11µm Active Pixel Sensor (APS) providing 6º HFOV at 30 Hz and 9.6 by 7.2 mm OLED with a 15 µm spot height and width and viewing distance of 2.35 cm. The system has 3.5 Magnification.

**Alternative 4 (b)** is an Increment II optically fused thermal and Digital Image Intensifier ($DI^2$) standalone sight with a 93.8 mm lens, and 640x480 17µm FPA providing 7 º HFOV at 30Hz. The $I^2$ component is a 75.3 mm lens, 1280x1024 11µm EBAPS providing 10 º HFOV at 30 Hz. This configuration merges two images from the thermal and $I^2$ displays through to the eye. The thermal display is 9.6 by 7.2 mm and the $I^2$ display is a 1.64 by 1.23 cm; both OLED(s) with a 25 µm spot height and width. The $I^2$ and thermal display viewing distance of 2.35 cm through the prism to limit parallax. The individual modalities and overall system have 3.5 Magnification. The optical fusion approach was originally developed to merge $I^2$ tube optical output with thermal images from a micro display. $I^2$ tubes directly project an optical image without the use of digital processing electronics. Developments in digital $I^2$ detectors have led to $I^2$ technologies that produce digital image outputs suitable for micro displays, and designers can now combine digital $I^2$ and thermal images electronically for a single display. Optical fusion is still an option for use with digital $I^2$ systems, and while it constrains selection of display optics, it does perform an optical registration and blending process that is complex and power intensive in digital processing. The optical fusion approach trades complexities in optical and mechanical design during the development process for system power and mission duration during operations. PM SSL guidance[10] was to focus on solutions that meet requirements, versus solutions that dovetail with longer-term development goals. Therefore, this report includes both optical fusion and digital fusion as alternatives.

**Alternative 5 (c)** is an Increment III digitally fused thermal and $DI^2$ standalone sight with a 93.8 mm lens, and 640x480 17µm FPA providing 7 º HFOV at 30Hz. The $I^2$ component is a 75.3 mm lens, 1280x1024 11µm EBAPS providing 10 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 1.64 by 1.23 cm OLED with a 25 µm spot height and width. The display viewing distance is 2.35 cm. The individual modalities and overall system have 3.5 Magnification.

**Alternative 6** is an Increment III Mid Wave Infrared (MWIR) standalone with a 150 mm lens, 320x240 29µm cooled detector providing 4 º HFOV at 60Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and viewing distance of 3.92 cm. The system has 3.83 Magnification.

**Alternative 7** is an Increment III Short Wave Infrared (SWIR) clip on sight with a 151.2 mm lens, 640x512 25µm FPA providing 5 º HFOV at 30Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 2.35 cm. The system has 3.59 Magnification. This alternative contains technologies that are Commercially-Off-The Shelf (COTS) but need to be ruggedized for military deployment.

**d. Sniper Rifle Alternatives**

Alternatives 2 and 3 were added to this weapon class from United States Army Infantry Center (USAIC) input to determine whether Mid Wave Infrared (MWIR) technology meets this study's Measures of Performance (MOP) and Measures of Effectiveness (MOE).

**Base System** is the AN/PAS – 13D V(3) Thermal Weapon Sight. It has a 99 mm lens, 640x480 25 µm Focal Plane Array (FPA) providing a 9 º Horizontal Field of View (HFOV), at a 30 Hertz

---

[10] Meeting Minutes [Word]. *Future Weapon Sight Technical Trade Study, Initial Meeting Minutes*; May 23, 2008.

(Hz) frame rate. The display has a 7.68 by 5.76 mm Liquid Crystal Display (LCD) with a 12 µm display spot height and width and viewing distance of 18.8 mm. The system has 2.55 Magnification.
**Alternative 1** is a standalone thermal sight in a currently funded program with a 65.1 mm lens, 640x480 17µm FPA providing 10 º HFOV at 30Hz. It has a 9.6 by 7.2 mm LCD with a 15 µm spot height and width and viewing distance of 2.35 cm. The system has 2.45 Magnification.
Alternative 2 and 3 are weapon sights with Mid Wave IR technology recommended by USAIC for analysis. These two alternatives are technologies currently under production by FLIR as the HISS 150 and 240. The 20 by 15cm size of the OLED was determined so that the display ratio is equivalent to the lens aperture of the day optic; to give proper magnification of the thermal image. (J. Hodge – FLIR, personal communication, March 25, 2009).

**Alternative 2** is a clip on Mid Wave Infrared (MWIR) thermal sight in a currently funded program with a 150 mm lens, 320x240 29µm cooled detector providing 4 º HFOV at 60Hz. It has a 20 by 15 cm OLED with a 0.32 mm spot height and width and viewing distance of 38.1 cm. The system has 8.34 Magnification.

**Alternative 3** is a clip on MWIR thermal sight in a currently funded program with a 240 mm lens, 320x240 29µm cooled detector providing 2 º HFOV at 60Hz. It has a 20 by 15 cm OLED with a 0.32 mm spot height and width and viewing distance of 38.1 cm. The system has 15 Magnification.

**Alternative 4** is an Increment I clip on thermal sight similar optimized to meet the requirements in this study. It has a 126 mm lens, 640x480 17µm FPA providing 5 º HFOV at 30Hz. It has a 7.2 $mm^2$ OLED with a 15 µm spot height and width and a viewing distance of 11.8 cm. The system has 1x Magnification. The technologies are readily available for production.

**Alternative 4 (b)** is similar to this alternative but it has a display viewing distance of 2.35 cm providing this variant a magnification of 5x, thereby making it a standalone configuration. A major constraint in this study is that clip – on systems shall have unity magnification.

**Alternative 5** is an Increment II clip on image intensifier with Electron Bombarded Active Pixel Sensor (EBAPS) technology and a 127.1 mm lens, 1280x1024 11µm Active Pixel Sensor (APS) providing 6º HFOV at 30 Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and a viewing distance of 8.28 cm. The system has 1x Magnification.

**Alternative 6 (b)** is an Increment II optically fused thermal and Digital Image Intensifier ($DI^2$) clip on sight. It is a 93.8 mm lens, 640x480 17µm FPA providing 7 º HFOV at 30Hz. The $I^2$ component is a 75.3 mm lens, 1280x1024 11µm EBAPS providing 10 º HFOV at 30 Hz. This configuration blends the two images from the thermal and $I^2$ displays to the eye. The thermal display is 9.6 by 7.2 mm and the I2 display is 1.61 by 1.21 cm; both are OLED(s) with a 25 µm spot height and width. The $I^2$ and thermal display viewing distance are both 8.28 cm through the prism to limit parallax. The individual modalities and overall system have 1x Magnification. The optical fusion approach was originally developed to merge $I^2$ tube optical output with thermal images from a microdisplay. $I^2$ tubes directly project an optical image without the use of digital processing electronics. Developments in digital $I^2$ detectors have led to $I^2$ technologies that produce digital image outputs suitable for micro displays, and designers can now combine digital $I^2$ and thermal images electronically for a single display. Optical fusion is still an option for use with digital $I^2$ systems, and while it constrains selection of display optics, it does perform an optical registration and blending process that is complex and power intensive in digital processing. The optical fusion approach trades complexities in optical and mechanical design during the development process for system power and mission duration during operations. PM SSL guidance[11] was to focus on solutions that meet requirements, versus solutions that dovetail with longer-term development goals. Therefore, this report includes both optical fusion and digital fusion as alternatives.

**Alternative 7 (c)** is an Increment III digitally fused thermal and $DI^2$ clip on sight with a 93.8 mm lens, 640x480 17µm FPA providing 7 º HFOV at 30Hz. The $I^2$ component is a 75.3 mm lens, 1280x1024 11µm EBAPS providing 10 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 1.61 by 1.21 cm OLED with a 25 µm spot height and width and viewing distance of 8.28 cm. The individual modalities and overall system have 1x Magnification.

**Alternative 8** is an Increment III Short Wave Infrared (SWIR) clip on sight with a 148.1 mm lens, 640x512 25µm FPA providing 6 º HFOV at 30Hz. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a display viewing distance of 8.33 cm. The system has 1x Magnification. This alternative contains technologies that are Commercially-Off-The Shelf (COTS) but need to be ruggedized for military deployment.

**NOTE: This study eliminated Alternative 9, 10, and 11 from the analysis and is not recommended as viable alternatives because their apertures are greater than 10.5 cm; maximum allotted by conversations with stakeholders. They are shown here to reiterate the increase in aperture when applying annular folded lens designs.**

**Alternative 9** is an Increment III thermal clip on similar to Alternative 4 but with annular folded optics. This system requires a 171 mm diameter lens as opposed to a 105 mm lens to capture the same amount of light. Like Alternative 4, it has a 126 mm focal length, 640x480 17µm FPA providing a 5 º HFOV at 30Hz and a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a viewing distance of 2.35 cm. The system has 1x Magnification.

**Alternative 10** is an Increment III clip on image intensifier with EBAPS similar to Alternative 5 but with annular folded optics technology. This system requires a 173 mm diameter lens as opposed to a 105 mm lens to capture the same amount of light. Like Alternative 5, it has a 127 mm focal length, 1280x1024 11µm Active Pixel Sensor (APS) providing 6 º HFOV at 30 Hz and a 9.6 by 7.2 mm OLED with a 15 µm spot height and width with a viewing distance of 8.28 cm. The system has 1x Magnification.

**Alternative 11** is an Increment III digitally fused thermal and $DI^2$ clip on similar to Alternative 7 (c) but with annular folded optics. This system requires a 127 mm diameter thermal and 173 mm $I^2$ lens to capture the same amount of light. It has a 640x480 17µm FPA providing 7 º HFOV at 30Hz and an $I^2$ 1280x1024 11µm EBAPS providing 6 º HFOV at 30 Hz. Digitally fused systems share the same display because a processor fuses the images prior to imaging. The display is a 1.61 by 1.21 cm OLED with a 25 µm spot height and width with a viewing distance of 8.28 cm. The individual modalities and overall system have 1x Magnification.

[11] Meeting Minutes [Word]. *Future Weapon Sight Technical Trade Study, Initial Meeting Minutes*; May 23, 2008.

**Alternative 12** is an Increment III image intensifier similar to Alternative 5 but with adaptive polymer optics. This system uses EBAPS technology with adaptive polymer optics to provide optical zoom from 1 to 2x. It has a 127.1 mm lens, 1280x1024 11µm providing a 6 ° HFOV at 1x and when actuated to 254 mm provides a 3 ° HFOV at 2x. It has a 9.6 by 7.2 mm OLED with a 15 µm spot height and width and display viewing distance of 8.28 cm.

**3. Objective.**

The Objective of this document is to identify mission capability gaps and record the results of the Analyses of Alternatives to meet the gaps with measurable metrics.

**4. Criteria.**

This study provides decision makers with analytical data to support development and procurement of technologies for small arms weapon sight systems.

Previous related documents were reviewed to identify performance parameters, and effectiveness metrics to determine the best alternatives within each weapon class. This analysis includes Measures of Performance (MOP) and Measures of Effectiveness (MOE) metrics to compare the alternatives.

Alternatives proposed in this study shall be limited to supporting ground combat forces for surveillance and close combat operations under all visibility conditions.

**5. Assumptions.**

This study provided the following assumptions for the deployment of each of the alternatives:

- The scenarios listed are consistent with the Mission Needs Statement (MNS)
- Capability gaps identified in the Small Arms Capabilities Based Assessment (SA CBA) will not fundamentally change in the operational environment
- Base Alternatives are acceptable for each weapon class
- Thermal Imagers image under Quarter Moon and Clear Starlight Conditions

The scenarios in this study were drafted from existing scenarios identified in the Family of Individual Optics (FOIO) Business Case Analysis (BCA).[12] The Measures of Performance (MOP), Measures of Effectiveness (MOE) and alternatives were not derived from the scenarios. The scenarios are primarily used to give context for the operational environment of each of the alternatives. For example, in Section 2.1 the Individual Rifle is used for the Urban Patrol Scenario; however, a soldier using an M249 Semi Automatic Weapon (SAW) could also provide support in the same scenario.
This study relied on the capability gaps identified by the SA CBA to develop the MOE in this study. Changing the capabilities during system deployment in the operational environment will refute the MOE results and ultimately the alternatives recommended for development and deployment.

[12] Program Manager Optics and Non-Lethal Weapon Systems Infantry Weapons Systems Directorate Marine Corps Systems Command (2008), *Family of Individual Optics (FOIO) Business Case Analysis (BCA).*

The Base Alternatives selected in each of the performance and effectiveness analysis sets the reference point and influences the results if the baseline system is changed. In this study the Thermal Weapon Sight (TWS) systems are the baseline for each of the weapon classes to include the sniper rifles. The magnitude of MOE and MOP improvements against a different Base Alternative, such as an Image Intensifier ($I^2$) system, will produce different results for each MOE and MOP metric.
The FWS SOW lists requirements for system range performance under specific illuminating and atmospheric conditions. Specifically, systems must meet requirements under clear day, dirty battlefield, adverse weather, quarter moon and clear starlight conditions. Only the SSCamIP, IINVD, and IICamIP have quarter moon and clear starlight as input parameters in their models; NVThermIP does not. The NVThermIP models imagers which sense emitted infrared radiation which is not affected by the illuminating conditions. This study shall apply the clear day range performance of thermal imagers to the quarter moon and clear starlight range requirements.

## 6. Recommendations

Individual Rifle Weapon Class

All recommended alternatives are clip-on designs that reduce weapon sight weight and allow the operator to maintain their day capability. The recommended option is a dual-band thermal/tube clip-on system similar in design to the ENVG (suitable for Increment II development). The NIR tube has low power and weight impacts, and offers good performance over the effective range of the Individual Rifle. The secondary option is the CNVD-T LWIR sight; the CNVD-T is suitable for Increment I fielding and provides excellent range performance in a light-weight package. Additional recommendations include a new digitally-fused LWIR/NIR system based on EBAPS technology (higher power than tube option) and a new small aperture LWIR optimized for weight and power. Detailed design information for each concept is in the body of the report.

| Individual Rifle Results | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **Alternative** | **Modality** | **Configuration** | **Range (norm)** | **HFOV (deg)** | **Aperture (cm)** | **Weight (lb)** | **Power (W)** | **Display Type** |
| Baseline | AN/PAS-13dV(1) | Single-Band | Standalone | 1.00 | 18.0 | 2.5 | 1.6 | 2.0 | LCD |
| #1 (Best) | LWIR<br>NIR (Tube) | Dual-Band<br>optically-fused | Clip-on | 2.09<br>1.09 | 7.0<br>14.0 | 3.3<br>5.3 | 2.5 | 2.0 | OLED |
| #2 | LWIR (CNVD-T) | Single-Band | Clip-on | 1.73 | 10.0 | 4.0 | 1.3 | 2.0 | OLED |
| #3 | LWIR<br>NIR (DI2) | Dual-Band<br>digitally fused | Clip-on | 2.05<br>1.28 | 7.0<br>13.0 | 3.3<br>5.0 | 2.6 | 3.2 | OLED |
| #4 | LWIR | Single-Band | Clip-on | 1.20 | 14.0 | 1.8 | 1.3 | 1.6 | OLED |

Light & Medium Machine Gun Weapon Class

All recommended alternatives are standalone designs based on limited rail-space. The recommended option is a new LWIR optimized for weight and power (Increment II). The single-band design provides the longest possible range performance in the light, low-power package. The second-best options are dual-band LWIR/NIR systems based on EBAPS technology, with an optically-fused version having power advantages over a digitally-fused version. The NIR channel offers effectiveness improvements (ability to image through glass at shorter ranges and ability to see NIR laser pointers) and has significantly shorter range performance than the baseline TWS. The heavy TWS AN/PAS-13D V(3) is a potential alternative for immediate Increment I production, if the significantly smaller FOV is acceptable.

| Light and Medium Machine Gun Results | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Alternative | Modality | Configuration | Range (norm) | HFOV (deg) | Aperture (cm) | Weight (lb) | Power (W) | Display Type |
| Baseline | AN/PAS-13dV(2) | Single-Band | Standalone | 1.00 | 18.0 | 4.3 | 2.8 | 3.0 | LCD |
| #1 (Best) | LWIR | Single-Band | Standalone | 1.14 | 18.0 | 2.8 | 1.9 | 1.9 | OLED |
| #2 | LWIR<br>NIR (DI2) | Dual-Band<br>optically-fused | Standalone | 1.14<br>0.84 | 18.0<br>10.0 | 2.8<br>7.9 | 2.6 | 2.5 | OLED |
| #3 | LWIR<br>NIR (DI2) | Dual-Band<br>digitally-fused | Standalone | 1.14<br>0.67 | 18.0<br>17.0 | 2.8<br>3.8 | 2.5 | 3.5 | OLED |
| #4 | AN/PAS-13dV(3) | Single-Band | Standalone | 1.92 | 9.0 | 9.0 | 3.5 | 3.0 | LCD |

Heavy Machine Gun Weapon Class

All recommended solutions are standalone designs since Heavy Machine Guns are currently fielded without day optics. The recommended option is a new cooled, MWIR system that provides excellent range performance at the cost of weight and power (cryocooler). The second-best option is a new LWIR design optimized for range performance given a maximum 10.5cm aperture. The LWIR system is lower power and lighter than the MWIR, but cannot effectively match MWIR range performance without moving to a much larger aperture. Additional options include LWIR/NIR systems based on EBAPS technology, with an optically-fused version having power advantages over a digitally-fused version. System design factors (available aperture and F/#) leads to fused LWIR channels with wider FOVs and shorter ranges than comparable single-band designs. The NIR channel offers effectiveness improvements (ability to image through glass at shorter ranges and ability to see NIR laser pointers) and has roughly half the range performance of the current Heavy TWS.

| Heavy Machine Gun Results | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Alternative | Modality | Configuration | Range (norm) | HFOV (deg) | Aperture (cm) | Weight (lb) | Power (W) | Display Type |
| Baseline | AN/PAS-13dV(3) | Single-Band | Standalone | 1.00 | 9.0 | 9.0 | 3.5 | 3.0 | LCD |
| #1 (Best) | MWIR | Single-Band | Standalone | 1.72 | 4.0 | 3.8 | 4.0 | 3.5 | OLED |
| #2 | LWIR | Single-Band | Standalone | 1.47 | 5.0 | 10.5 | 3.3 | 1.9 | OLED |
| #3 | LWIR<br>NIR (DI2) | Dual-Band<br>optically-fused | Standalone | 1.21<br>0.52 | 7.0<br>10.0 | 7.8<br>2.7 | 2.9 | 2.5 | OLED |
| #4 | LWIR<br>NIR (DI2) | Dual-Band<br>digitally-fused | Standalone | 1.21<br>0.52 | 7.0<br>10.0 | 7.8<br>2.7 | 2.9 | 3.5 | OLED |

Sniper Rifle Weapon Class

The recommended options for the Sniper Rifle are a trade-off between clip-on and standalone design concepts. Clip-on performance is constrained by the interface to the Leopold day optic, and standalone designs require the Sniper to remove the day optic (not desirable). The HISS 240mm MWIR sight is suitable for Increment I production if the reduced FOV (2 degrees) is acceptable to the user. Otherwise, a new MWIR clip-on (possibly a modified HISS) is the best choice. Dual-band clip-on systems (shown for reference in the table below) are unable to provide adequate range performance, with the thermal channels reaching roughly 85% of the

Baseline, and the NIR channels reaching roughly 35% of the baseline. The Heavy Machine Gun standalone alternatives apply equally to the Sniper Rifle results, with a new MWIR and new LWIR as recommended results. Standalone dual-band LWIR/NIR systems offer a significant capability if targets will be engaged within the range of the NIR sensor.

| Sniper Rifle Results | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Alternative | Modality | Configuration | Range (norm) | HFOV (deg) | Aperture (cm) | Weight (lb) | Power (W) | Display Type |
| Baseline | AN/PAS-13dV(3) | Single-Band | Standalone | 1.00 | 9.0 | 9.0 | 3.5 | 3.0 | LCD |
| #1 (Best) | MWIR (HISS 240) | Single-Band | Clip-on | 1.93 | 2.0 | 6.0 | 4.0 | 3.0 | OLED |
| #2 | MWIR | Single-Band | Standalone | 1.72 | 4.0 | 3.8 | 4.0 | 3.5 | OLED |
| #3 | LWIR | Single-Band | Standalone | 1.47 | 5.0 | 10.5 | 3.3 | 1.9 | OLED |
| #4 | LWIR<br>NIR (DI2) | Dual-Band<br>optically-fused | Standalone | 1.21<br>0.52 | 7.0<br>10.0 | 7.8<br>2.7 | 2.9 | 2.5 | OLED |
| #5 | LWIR<br>NIR (DI2) | Dual-Band<br>digitally-fused | Standalone | 1.21<br>0.52 | 7.0<br>10.0 | 7.8<br>2.7 | 2.9 | 3.5 | OLED |
| reference | LWIR<br>NIR (DI2) | Dual-Band<br>optically-fused | Clip-on | 0.86<br>0.35 | 7.0<br>10.0 | 7.8<br>2.7 | 2.8 | 2.5 | OLED |
| reference | LWIR<br>NIR (DI2) | Dual-Band<br>digitally-fused | Clip-on | 0.86<br>0.35 | 7.0<br>10.0 | 7.8<br>2.7 | 2.8 | 3.6 | OLED |

f. Conclusion

The study identified three systems suitable for Increment I production (CNVD-T, heavy TWS, and HISS 240mm). Increment II development recommendations focus on new single-band weapon sights (LWIR clip-on for the Individual Rifle; LWIR standalone for the LMG/MMG; MWIR standalone for HMG; and modified HISS MWIR for Sniper Rifle). In addition, dual-band concepts ranked well across the board, and Increment II development should include digitally fused sights that can be enhanced during Increment III. Optically-fused dual-band concepts provide low-power alternatives, and Increment II development is suitable as an interim solution until advanced digital fusion can be developed in Increment III.

| Weapon Class | Increment I | Increment II | Increment III |
|---|---|---|---|
| Individual Rifle | CNVD-T clip-on | LWIR clip-on<br>LWIR/NIR tube clip-on | LWIR/NIR adv fusion |
| Light Machine Gun<br>Medium Machine Gun | AN/PAS - 13dV(3) | LWIR standalone<br>LWIR/NIR EBAPS stand | LWIR/NIR adv fusion |
| Heavy Machine Gun | ----- | MWIR standalone<br>LWIR standalone<br>LWIR/NIR EBAPS stand | LWIR/NIR adv fusion |
| Sniper Rifle | HISS (240mm) clip-on | modified HISS clip-on<br>MWIR standalone<br>LWIR standalone<br>LWIR/NIR EBAPS stand | LWIR/NIR adv fusion |

Appendix E: – System Required Capabilities Tables

## Table 6.1 – KPPs

| Tier 1 & 2 JCA | Key Performance Parameter | Development Threshold | Development Objective |
|---|---|---|---|
| | KPP1 | | |
| Tier 1<br>Force Application<br>Tier 2<br>Engage<br>Maneuver | Physical Compatibility | The FWSI shall mount using a MIL-STD 1913 Integrated rail system in line with the M150 Rifle Combat Optic (RCO) and the M68 Close Combat Optic (CCO) on individual Soldier weapons (M16A2,A4/M4/M249 M136 AT4 and M141 BDM.). It will also have the ability to operate in a standalone configuration. This design will include a Rapid Target Acquisition (RTA) which displays the reticle and a portion of a thermal image in the HMD. The RTA design will include a wired interface between the FWSI and HMD and the associated signal processing electronics.<br><br>The FWSCS shall be a standalone limited visibility/night optic that mounts on the MIL-STD-1913 Integrated Rail of the M240, M2, MK19, & MK47. It shall be configured as a fused sight with a Low Light Level (LLL) and a thermal fused sensor.<br><br>The FWSCS shall have a Head Mounted Display (HMD) Wired<br><br>The FWS all variants will allow for a zeroed FWS to be repeatedly attached with a constant return to zero within .50mrad.<br><br>The FWS Sniper shall mount on the MIL-STD-1913 Integrated Rail of the M110, M24 and M107 in-line with the Leupold MK4 4.5-14X day optic, and shall be used in the stand-alone mode. It shall not interfere in any way with the normal operation or inherent accuracy of the host weapon. | The FWSI shall mount using a MIL-STD 1913 Integrated rail system in line with the M150 Rifle Combat Optic (RCO) and the M68 Close Combat Optic (CCO) on individual Soldier weapons (M14, MK16, MK17 Rifle) (O)<br><br>FWSI shall have a range finding capability to the effective ranges of the weapons described in this CDD within of +/- 5 meter (O)<br><br>The RTA design will include a wireless interface between the FWSI and HMD and the associated signal processing electronics.<br><br>T=O<br><br>FWSCS shall have a range finding capability of +/- 5 meter (O) and wind measurement +/- 0.23 meters per second (O).<br><br>The FWSCS shall have a Head Mounted Display (HMD) Wireless.<br><br>The FWSCS shall have the ability to interface with a laser range finding device and receive range information as inputs to ballistic equations embedded within the FWSCS and shall have a range finding capability of +/- 1 meter (O) and wind measurement +/- 0.23 meters per second (O).<br><br>The FWS all variants will allow for a zeroed FWS to be repeatedly attached with a constant return to zero within .25mrad<br><br>T=O |
| | KPP 2 | | |
| Tier 1<br>Battlespace Awareness<br>Tier 2<br>Collection | Target Detection | The FWS must provide the user the ability to detect/recognize man-sized targets during night and limited visibility scenarios, appropriate magnification is needed to achieve the required probability of recognition:<br>During night<br><br>FWS Individual | During night<br><br>FWS Individual |

| Tier 1 & 2 JCA | Key Performance Parameter | Development Threshold | Development Objective |
|---|---|---|---|
| | | >0.7 P/r 960m<br><br>FWS Crew Served<br>>0.7 P/r 2400m<br><br>FWS Sniper<br>>0.7 P/r 1800m<br><br>Smoke or other obscurants<br><br>FWS Individual<br>>0.9 P/r 300m<br><br>FWS Crew Served<br>>0.9 P/r 500m<br><br>FWS Sniper<br>>0.9 P/r 600m | >0.7 P/r 1000m<br><br>FWS Crew Served<br>>0.7 P/r 2600m<br><br>FWS Sniper<br>>0.7 P/r 2200m<br><br>Smoke or other obscurants<br><br>FWS Individual<br>>0.9 P/r 480m<br><br>FWS Crew Served<br>>0.9 P/r 600m<br><br>FWS Sniper<br>>0.9 P/r 800m |
| | KPP 3 | | |
| Tier 1<br>Battlespace Awarness<br>Tier 2<br>Collection | Field Of View (FOV) | The FWS shall provide the following minimum FOV in a stand-alone mode.<br><br>FWS Individual<br>= 18.0 Deg FOV<br><br>FWS Crew Served<br>= 9.0 Deg FOV<br><br>FWS Sniper<br>= 4.0 Deg FOV | FWS Individual<br>= 26.0 Deg FOV<br><br>FWS Crew Served<br>= 18.0 Deg FOV<br><br>FWS Sniper<br>= 9.0 Deg FOV |
| | KPP 4 | | |
| Tier 1<br>Force Application<br>Tier 2<br>Maneuver | Weight | The total system weight of the FWS Individual with batteries, without carrying case cannot exceed 1.75 lbs. RTA processor shall weigh less than 2.0 lbs excluding sight and HMD.<br><br>The total system weight of the FWS CS with batteries, without carrying case cannot exceed 3.50 lbs.<br><br>The total system weight of the FWS Sniper with batteries, without carrying case cannot exceed 2.5 lbs. | The total system weight of the FWSI and batteries to include additional loaded battery cassette, without carrying case cannot exceed 1.75 lbs (O). RTA processor shall weigh less than 1.5 lbs (O) excluding sight and HMD (O).<br><br>The total system weight of the FWS CS with batteries, without carrying case cannot exceed<br>3.25 lbs.<br><br>The total system weight of the FWSS and batteries to include additional loaded battery cassette, without carrying case cannot exceed 1.75 lbs (O). |
| | KPP 5 | | |
| Tier 1<br>Force Application<br>Tier 2<br>Maneuver | Power | All FWS variants shall be operable by internal battery power in all weather conditions for at least 7 hours full-on with one battery change. All FWS variants shall have a low battery indicator that alerts the operator when approx 15 minutes of battery life | The FWS all variants shall be operable by internal battery power in continuous power-on mode in all weather conditions for at least 7 hours with one set of batteries. All FWS variants shall have a battery life indicator that provides battery health through the battery life. |

| Tier 1 & 2 JCA | Key Performance Parameter | Development Threshold | Development Objective |
|---|---|---|---|
| | | remains. | |

*Return to Paragraph 6, KPPs*

**Table 6.2 – KSAs**

| Tier 1 & 2 JCA | Key System Attributes | Development Threshold | Development Objective |
|---|---|---|---|
| | KSA 1 | | |
| Tier 1<br>Force Application<br>Tier 2<br>Engage | Focus adjustments | The FWSI will have a minimum focus of 5 meter (T)<br><br>The FWSCS and FWSS shall have a focus adjustment range of 10 meters to infinity. | The FWSI will have a minimum focus of 1 meter (O).<br><br>The FWSCS and FWSS shall have a focus adjustment range of 5 meters to infinity. |
| | KSA 2 | | |
| Tier 1<br>Force Application<br>Tier 2<br>Maneuver | Survivability | The FWS all variants shall not emit noise that is detectable by the average human ear at a distance beyond five meters in any direction during use.<br><br>When the user is wearing protective eyewear or goggles, the FWS all variants shall not emit stray light detectable by an unaided eye beyond 5 meters. | T=O |
| | KSA 3 | | |
| Tier 1<br>Logistics<br>Tier 2<br>Deployment & Distribution<br>Maintain | Ownership Cost | The total cost of FWS (to include: basic unit, carrying case, ancillary equipment and sustainment costs) shall not exceed (BY11 $).<br><br>FWS Individual<br>= $23,020 Per Unit<br><br>FWS Crew Served<br>= $23,600 Per Unit<br><br>FWS Sniper<br>= $23,150 Per Unit | The total cost of FWS (to include: basic unit, carrying case, ancillary equipment and sustainment costs) shall not exceed (BY11 $).<br><br>FWS Individual<br>= $25,320 Per Unit<br><br>FWS Crew Served<br>= $25,960 Per Unit<br><br>FWS Sniper<br>= $25,460 Per Unit |
| | KSA 4 | | |
| Tier 1<br>Logistics<br>Tier 2<br>Maintain | Operational Availability | The FWS will be no less than 81 percent measured continuously over the 8-day wartime usage period (consisting of two aggregate 96-hour wartime scenarios) depicted in the | The FWS will be no less than 90 percent measured continuously over the 8-day wartime usage period (consisting of two aggregate 96-hour wartime scenarios) depicted in the Operational Mode |

| | | | |
|---|---|---|---|
| | | Operational Mode Summary/Mission Profile (OMS/MP). The assessment of $A_o$ shall be consistent with equipment usage factors described in the OMS/MP, the maintenance support concept implemented for the FWS, and repair cycle Administrative and Logistics Delay Time (ALDT) typical of that experienced in the operational/field environment, where System Abort failure events as described by the FWS Reliability Failure Definition and Scoring Criteria (FDSC) are the primary indicator of equipment not ready/available conditions. | Summary/Mission Profile (OMS/MP). The assessment of $A_o$ shall be consistent with equipment usage factors described in the OMS/MP, the maintenance support concept implemented for the FWS, and repair cycle Administrative and Logistics Delay Time (ALDT) typical of that experienced in the operational/field environment, where System Abort failure events as described by the FWS Reliability Failure Definition and Scoring Criteria (FDSC) are the primary indicator of equipment not ready/available conditions. |
| | **KSA 5** | | |
| Tier 1 Logistics Tier 2 Maintain | Reliability | The FWS will achieve/demonstrate an operational (probabilistic) reliability of not less than .90 (T), with regard to successfully completing the 96 hour wartime usage cycle depicted by the OMS/MP without incurring a System Abort failure event as defined in the Reliability Failure Definition and Scoring Criteria (FDSC) The FWS operational reliability required to complete the 96-hour scenario without incurring an EFF of any severity will be no less than .84 (T) | The FWS will achieve/demonstrate an operational (probabilistic) reliability of not less than .95 with regard to successfully completing the 96 hour wartime usage cycle depicted by the OMS/MP without incurring a System Abort failure event as defined in the Reliability Failure Definition and Scoring Criteria (FDSC) The FWS operational reliability required to complete the 96-hour scenario without incurring an EFF of any severity will be no less than .92 (T) |

*Return to Paragraph 6, KSAs*

**Table 6.3 – Additional Performance Attributes**

| Attribute | Development Threshold | Development Objective |
|---|---|---|
| **Attribute 1** | | |
| Carrying Case | All FWS variants shall have a soft water resistant carrying case that shall attach to individual load bearing equipment and a hard transit case. | T=O |
| **Attribute 2** | | |
| Battery Removal | The battery cassette and batteries must be easily replaced without using tools, without removing the sight from the weapon, and without affecting zeroing/sight alignment. | T=O |
| **Attribute 3** | | |
| Compatibility with Personal Protective Equipment (PPE) | All FWS variants shall be capable of being operated while wearing issued personnel protective equipment. *Note: PPE includes equipment such as the Light Weight Helmet, MICH Helmet, Modular Tactical Vest, Gloves (cold weather, NOMEX) protective eyewear, and MOPP IV gear.* | T=O |
| **Attribute 4** | | |
| Reticle | Each FWS will have internal selectable reticles that are compatible with each supported weapon. | T=O |

| | | |
|---|---|---|
| **Attribute 5** | | |
| Gain/Brightness | All FWS variants should allow the operator to control and adjust the image quality manually. | T=O |
| **Attribute 6** | | |
| Durability | The FWS materials shall protect it from corrosion in all environments and weather conditions, including marine, high humidity, rain and desert conditions | freezing precipitation (O) |
| **Attribute 7** | | |
| Survivability | All FWS variants shall be provided lens covers with lanyards. An additional attachment point for a security lanyard shall be provided. | T=O |
| **Attribute 8** | | |
| Remote viewing port | All FWS variants shall have a Input/Output (I/O) port. This I/O port will facilitate the heads-up display, remote viewing and external power. | T=O |
| **Attribute 9** | | |
| Operational Readiness | Over the full operating temperature range the FWS shall provide a recognizable image within 120 seconds after FWS power is applied and within 3 seconds after being switched from stand-by to full on. | T=O |
| **Attribute 10** | | |
| Weapon boresight retention | The FWS, when mated to any of the weapons specified shall retain bore sight over all environmental conditions. | T=O |
| **Attribute 11** | | |
| Remote control | All FWS, shall have a wired remote control that has at least 3 buttons to control the following Polarity, FOV, and Calibration | The FWS, shall have a wireless remote control that has at least 3 buttons to control the following, Polarity, FOV, and Calibration |

*Return to Paragraph 6, Additional Performance Attributes*